

OLIVER'S NOTES IN MATHEMATICS AND PHYSICS

1. Introductory Abstract Algebra

Groups, Rings, Fields, and Modules



Oliver Kahng

First Edition

2026

Preface

This text serves as a first rigorous introduction to abstract algebra. Readers are assumed to have knowledge of elementary linear algebra, as well as some experience in writing proofs. No prior experience with abstract algebra is needed to read this book. The pace and content of this book are very similar to Michael Artin's treatment of the same topic, though with some of the linear algebra sections removed.

The book begins with a few chapters on the fundamentals of group theory and representation theory. Afterwards, the focus shifts toward rings, and then modules. Finally, the text covers fields and Galois theory. There is an optional appendix on homological algebra for the interested reader.

I would like to thank Michael Artin, Serge Lang, and David Dummit and Richard Foote for writing the texts in algebra that influenced the content and exercises found in this book. More information can be found in the Further Reading section.

How to read this book

Definitions are highlighted in blush boxes; examples appear in pale lavender; and remarks use a quieter neutral frame. Important results are set as theorems or propositions, followed by proofs in the main text. Exercises at the end of each chapter are intended to be part of the exposition rather than an afterthought; the reader should complete all of the exercises, as there are not many of them. There are also problems scattered throughout each chapter; the reader should also complete all of those in order to gain the most complete understanding of the subject. Problems are generally easier and can even be done in the reader's head; exercises are generally harder and require more thought.

It is possible, and recommended, to fit the entire content of this book into one semester, with some room for additional topics. Alternatively, this book could be used as the basis for a very leisurely two semester course, alongside supplementary texts provided by the professor as necessary.

Oliver Kahng
2026

Contents

Preface	iii
1 Introduction to Groups	1
1.1 Groups	1
1.2 Subgroups	5
1.3 Homomorphisms	9
1.4 Equivalence Relations	13
1.5 Cosets	16
1.6 Modular Arithmetic	19
1.7 Product and Quotient Groups	21
1.8 Exercises	24
2 Symmetry	25
2.1 Isometries	25
2.2 Cyclic and Dihedral Groups	30
2.3 Crystallographic Groups	34
2.4 Group Actions	40
2.5 The Orbit-Stabilizer Theorem	43
2.6 Exercises	44
3 Introduction to Rings	46
3.1 Rings	46
Further Reading	47

CHAPTER 1

Introduction to Groups

“In these days the angel of topology and the devil of abstract algebra fight for the soul of each individual mathematical domain.”

Hermann Weyl

1.1 Groups

To understand groups, we must first understand what a law of composition is. Luckily, you are already familiar with examples of laws of composition from previous mathematics. For example, the addition operation may take in two integers, with the result being another integer. For example, $4 + 3 = 7$. In this case, we have that 4 and 3 are our inputs and 7 is our output. We say, therefore, that the set $\mathbb{Z}$ (of integers) is equipped with the addition operation $+$, where

$$+ : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \tag{1.1}$$

and

$$(4, 3) \mapsto 7 \tag{1.2}$$

under the addition operation.

More generally, we say that a law of composition is defined below.

Definition 1.1 — Law of Composition

Suppose that we have a set S . A **law of composition** on a set S is defined to be any mapping

$$\circ : S \times S \rightarrow S \tag{1.3}$$

Recall that $S \times S$ denotes the **Cartesian product** of S with itself, or the set of ordered pairs (s, t) , where $s, t \in S$.

It is known that some laws of composition (but not all of them) are associative. Consider the addition operation from before. Recall that $(1 + 1) + 2 = 4 = 1 + (1 + 2)$. We observe that for any triplet of numbers added together, it does not matter how we group them, for it results in the same answer nevertheless. This property is known as associativity.

Definition 1.2 — Associativity

A law of composition $\circ$ is defined to be **associative** if different groupings do not change the final result; that is,

$$(a \circ b) \circ c = a \circ (b \circ c) \quad (1.4)$$

Problem 1.1

Can you think of a law of composition for any set that is not associative?

It is, in fact, possible to define, given an associative law of composition, a product of n elements in the set based on the law of composition itself. We provide further information in the proposition below.

Proposition 1.1. *Suppose there exists an associative law of composition $\circ$ on a set S . We can uniquely define, where $n \in \mathbb{N}$, a product $[a_1 \cdots a_n]$ of n elements $a_1, \dots, a_n \in S$, such that:*

- i. $[a] = a$,
- ii. $[ab] = a \circ b$, and
- iii. For any $i \in \mathbb{N}$ such that $1 \leq i < n$, we must have $[a_1 \cdots a_n] = [a_1 \cdots a_i] \circ [a_{i+1} \cdots a_n]$.

Proof. The objective of this proof is to show uniqueness; that is, there is only *one* way to define a product in such a way. It is clear that we have uniquely defined the product for 1 and 2 elements, as there is no other way to define another such product such that (i), (ii), and (iii) hold. We use induction to prove uniqueness for $n > 2$. Suppose the product is uniquely defined for all $r \leq n - 1$ and satisfies all three properties. Define

$$[a_1 \cdots a_n] = [a_1 \cdots a_{n-1}] \circ [a_n] \quad (1.5)$$

As both terms on the right-hand side are defined uniquely, it follows that the left-hand side is defined uniquely. We now must show that the product of n elements satisfies condition (iii). We have:

$$[a_1 \cdots a_n] = [a_1 \cdots a_{n-1}] \circ [a_n] \quad (1.6)$$

Since the product of $n - 1$ elements is given to satisfy condition three, we have

$$[a_1 \cdots a_{n-1}] \circ [a_n] = [a_1 \cdots a_i] \circ [a_{i+1} \cdots a_{n-1}] \circ [a_n] \quad (1.7)$$

Furthermore, by condition (iii) once more, we may state that $[a_{i+1} \cdots a_{n-1}] \circ [a_n] = [a_{i+1} \cdots a_n]$, so

$$\begin{aligned} [a_1 \cdots a_i] \circ [a_{i+1} \cdots a_{n-1}] \circ [a_n] &= [a_1 \cdots a_i] \circ [a_{i+1} \cdots a_n] \\ \implies [a_1 \cdots a_n] &= [a_1 \cdots a_i] \circ [a_{i+1} \cdots a_n] \end{aligned} \quad (1.8)$$

□

At last we come to the definition of a group. The group is a very versatile mathematical object. Certain sets of numbers are groups. Sets of symmetries of objects may also be groups. The group as a mathematical object contains four fundamental properties: closure, associativity, identity, and inverse properties. **Closure** essentially means that if you take two elements and compose them (by

the given law of composition), the result must be an element of the group as well. Associativity is already known. The other two properties are defined as part of the definition of a group below.

Definition 1.3 — Group

A **group** is defined to be a set G equipped with a law of composition $\circ : G \times G \rightarrow G$, such that the following conditions hold:

- i. The law of composition is *associative*,
- ii. There exists some identity $1 \in G$, such that $1 \circ a = a \circ 1 = a$ for all $a \in G$, and
- iii. Every element $a \in G$ has an inverse $a^{-1} \in G$ such that $a \circ a^{-1} = a^{-1} \circ a = 1$.

Note that the closure is not included in the list of properties in the definition.

Problem 1.2

Why is it that the closure of a group is assumed based on the definition of a law of composition?

There is a special kind of group, one where the law of composition is commutative as well as associative. This group is known as an abelian group.

Definition 1.4 — Abelian Group

If the law of composition associated with a group G is commutative as well, that is, $a \circ b = b \circ a$ for all $a, b \in G$, then the group G is known as an **abelian group**.

Many familiar groups are abelian. For example, the set of integers $\mathbb{Z}$ under the law of composition, or operation, of addition, is an abelian group; recall the commutative property of addition: $a + b = b + a$ for all $a, b \in \mathbb{Z}$. One can verify that the set of integers with the operation of addition is a group through examining the three properties mentioned in the definition of a group.

One may also be concerned with the order of a group, or the *number of elements* the group has. A group may contain an infinite number of elements (like $\mathbb{Z}$) or a finite number of elements.

Definition 1.5 — Order

The **order** of a group G is simply the number of elements the group contains.

One of the most fundamental laws in group theory is the Cancellation Law. The theorem and proof are stated in full below.

Theorem 1.2 (Cancellation Law). Suppose that $a, b, c \in G$, and G is a group. Then, if $a \circ b = a \circ c$ or $b \circ a = c \circ a$, it must follow that $b = c$.

Proof. Suppose we are given that $a \circ b = a \circ c$. It must follow that

$$a^{-1} \circ (a \circ b) = a^{-1} \circ (a \circ c) \quad (1.9)$$

since the two expressions in parentheses are equivalent. Furthermore, by associativity, we have

$$(a^{-1} \circ a) \circ b = (a^{-1} \circ a) \circ c \quad (1.10)$$

Note that $a^{-1} \circ a = 1$, where 1 is the identity, so

$$1 \circ b = 1 \circ c \quad (1.11)$$

Since $1 \circ x = x$ for all $x \in G$, we therefore have

$$b = c \quad (1.12)$$

which is our desired result. The proof where $b \circ a = c \circ a$ will be left to the reader. $\square$

Problem 1.3

Verify the Cancellation Law for the other case; that is, given $b \circ a = c \circ a$, show $b = c$.

We now come to some examples of groups that may be of interest.

Example: The General Linear Group

The **general linear group**, which we denote as GL_n , is the set of all *invertible* $n \times n$ matrices (these matrices may take real or complex values, depending on which kind of general linear group we are referring to), equipped with the operation of matrix multiplication. Consider that matrix multiplication is associative. Furthermore, the $n \times n$ identity matrix

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

is invertible, and therefore serves as the identity element of GL_n . Furthermore, since every matrix $A \in GL_n$ is invertible, there must exist some A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n \quad (1.13)$$

It is clear from the equation above that A^{-1} is also invertible, and therefore is an element of GL_n . Hence GL_n satisfies all three properties of a group.

Problem 1.4

Is GL_n an abelian group?

Example: The Symmetric Group

The **symmetric group** S_n , is the set of permutations of the set of indices $\{1, \dots, n\}$. That is, it is the set of different arrangements of the numbers 1 to n , where each number only shows up once in each arrangement. For example, we have that S_3 would be $\{123, 132, 213, 231, 312, 321\}$. The operation is the combination of permutations. Consider each permutation as a map from $\{1, \dots, n\}$ to $\{1, \dots, n\}$. The operation in S_n is simply the composition of these maps. For example, 123 maps 1 to 1, 2 to 2, and 3 to 3. 321 maps 1 to 3,

2 to 2, and 3 to 1. 231 maps 1 to 2, 2 to 3, and 3 to 1. Therefore, if we have $321 \circ 123$, we will have 1 map to 1 by 123, which then maps to 3 by 321. 2 maps to 2, and then maps to 2. 3 maps to 3, which subsequently maps to 1. Hence, $321 \circ 123 = 321$. As it turns out, we can actually express S_3 in the form $\{1, x, x^2, y, xy, x^2y\}$, with the following relations defining the group:

$$x^3 = x \circ x \circ x = 1 \quad (1.14)$$

$$y^2 = y \circ y = 1 \quad (1.15)$$

$$yx = y \circ x = x^2y = x \circ x \circ y \quad (1.16)$$

Note that we are starting to drop the $\circ$ symbol for convenience. We also use x^n to denote $x \circ \cdots \circ x$, where x is found n times in the expression.

Problem 1.5

What is the order of S_4 ?

1.2 Subgroups

We are already familiar with the fact that $\mathbb{Z}$ is a group under addition. However, notice that the set of even numbers, which shall be denoted as $2\mathbb{Z}$, is also a group under addition, and a subset of $\mathbb{Z}$ as well. If a subset of a group is also a group under the same law of composition, then we call that subset a **subgroup**. More rigorously, subgroups must fulfill the following conditions.

Definition 1.6 — Subgroup

We define a **subgroup** H of a group G to be a subset of G such that the following conditions hold:

- i. **Closure:** If $a, b \in H$, then $ab \in H$,
- ii. **Identity:** The identity 1 of G is in H as well,
- iii. **Inverse:** If $a \in H$, then the inverse $a^{-1} \in H$ as well.

Notice that the *associativity* property is not listed above. This is because such a property is naturally inherited from the group that the subgroup is a subset of. What that means is that associativity holds for all triplets of elements in the group; therefore, since all elements in the subgroup are elements of the group, associativity must hold for all triplets of elements in the subgroup as well. Therefore, there is no need to verify that the associativity property holds for a subset of a group equipped with the same law of composition as its superset.

Below are a couple of varying examples of subgroups. The reader may choose to verify that they really are subgroups if they so desire.

Example: The Circle Group

The **circle group** S^1 is the set of complex numbers where the absolute value, or **modulus**, of each element is 1. It is a subgroup of the nonzero complex numbers $\mathbb{C}^\times$ under the operation

of multiplication.

Example: The Special Linear Group

The **special linear group**, denoted by SL_n , is the set of $n \times n$ matrices with determinant equal to 1; the equipped operation is matrix multiplication. Since any matrix with nonzero determinant is invertible, it is clear that SL_n is a subgroup of the general linear group GL_n .

Of particular interest are subgroups of $\mathbb{Z}$, the set of integers. Such subgroups naturally take one of two forms: either the trivial group consisting only of the element 0, or a group which we denote by $\mathbb{Z}a$, which is defined below. We shall prove this claim rigorously as well.

Theorem 1.3. *Suppose we have a subgroup S of the integers $\mathbb{Z}$ under addition. S must either be the trivial subgroup $\{0\}$ containing only the identity (in the case of $\mathbb{Z}$, 0 is the identity), or it must take the form $\mathbb{Z}a = \{n \in \mathbb{Z} \mid n = ka, \text{ where } k \in \mathbb{Z}\}$, where a is defined to be the smallest positive integer in S .*

Proof. Suppose S is a subgroup of $\mathbb{Z}$. It is necessary for S to contain the identity 0, by the definition of a subgroup. If there are no other elements in S , then S is the trivial subgroup. We will show that otherwise, $S = \mathbb{Z}a$ for some a . If S is not trivial, then S must contain an element n other than 0; note that either n or $-n$ is positive, and by the third property of a subgroup, S must contain both. Therefore, S contains a positive element, and so there must be a smallest positive integer in S . Let a be such a number. We will show that $S = \mathbb{Z}a$ by showing each is a subset of the other. Suppose we have an arbitrary $ka \in \mathbb{Z}a$, where $k \in \mathbb{N}$. Since k is positive, $ka = a + a + \cdots + a$, where a is repeated k times. $a + a$ must be in S , as a is in S , and by the property of closure. It follows by induction that $ka \in S$ as well. Meanwhile, $-ka \in S$ by the inverse property of a subgroup. Of course, $0a = 0 \in S$ as well. Therefore, $\mathbb{Z}a \subset S$, as every element in $\mathbb{Z}a$ is contained in S . Furthermore, suppose $n \in S$. Since n is an integer, we may write $n = qa + r$ for some $q, r \in \mathbb{Z}$, and $0 \leq r < a$. Note that $qa \in S$ as $\mathbb{Z}a \subset S$, and n is in S . By closure, we must have that $n - qa = r \in S$ as well. Recall that a is supposed to be the smallest positive integer in S , but $0 \leq r < a$. Therefore, r cannot be a positive integer, and must be 0. So, $n = qa + 0 = qa$, and therefore $n \in \mathbb{Z}a$. Hence, since any element of S is an element of $\mathbb{Z}a$, we have $S \subset \mathbb{Z}a$. Since $\mathbb{Z}a \subset S$ and $S \subset \mathbb{Z}a$, it follows necessarily that $S = \mathbb{Z}a$. Therefore, the claim has been verified. $\square$

The greatest common divisor, also known as the greatest common factor, is surely familiar to the reader. We shall define it here differently; specifically, the greatest common divisor of a and b is defined to be the integer of the subgroup $\mathbb{Z}a + \mathbb{Z}b$, which shall be defined in the proposition below. Our goal is to show that the familiar properties of the greatest common divisor arise naturally from this definition.

Proposition 1.4. *Suppose $a, b \in \mathbb{Z}$, where a and b are not both zero, and d is their greatest common divisor (GCD), which is defined to be the positive integer that generates the subgroup*

$$\mathbb{Z}d = \mathbb{Z}a + \mathbb{Z}b = \{n \in \mathbb{Z} \mid n = ra + sb, \text{ where } r, s \in \mathbb{Z}\} \quad (1.17)$$

These properties follow:

- i. *There exist $r, s \in \mathbb{Z}$ such that $d = ra + sb$,*
- ii. *d divides a and b , and*

iii. If c divides a and b , c also divides d .

Proof. Since $d \in \mathbb{Z}a + \mathbb{Z}b$, it follows that $d = ra + sb$ for some $r, s \in \mathbb{Z}$. Hence the first property is true. Furthermore, since $a, b \in \mathbb{Z}d$, that means $a = kd$ for some $k \in \mathbb{Z}$, and likewise for b ; therefore, d divides both a and b . The last property will be left to the reader to verify. $\square$

Problem 1.6

Verify property (iii) of the above proposition. *Hint:* Use the fact that $d = ra + sb$ for some $r, s \in \mathbb{Z}$.

There is an interesting and noteworthy method of calculating the greatest common divisor known as the Euclidean Algorithm. It is named after the famous mathematician Euclid, who described the algorithm for computing the greatest common divisor in his book *Elements*. It reduces the task of computing the greatest common divisor to simple division calculations, as the reader may see below.

Definition 1.7 — Euclidean Algorithm

The **Euclidean algorithm** is a method that allows one to quickly find the greatest common divisor of two integers through division. The process is simple.

1. Suppose we wish to find the GCD of a and b , where $a \geq b$.
2. Divide a by b , and find the remainder; that is, find r_1 such that $a = bq + r_1$ where $0 \leq r_1 < b$.
3. Now, divide b by r_1 , and find the remainder of that; that is, find r_2 such that $b = r_1q + r_2$.
4. Repeat this process iteratively until we have some $r_n = 0$.
5. It follows that r_{n-1} is the GCD of a and b .

It follows very shortly from the properties of the GCD that we may be able to consider **relatively prime** numbers, a pair of numbers where their greatest common divisor is equal to 1. Observe the following corollary:

Corollary 1.5. We must have that $a, b \in \mathbb{Z}$ are **relatively prime** (that is, their GCD is 1) if and only if there exist $r, s \in \mathbb{Z}$ such that $ra + sb = 1$.

Problem 1.7

Verify this corollary. Be sure to prove in both directions!

Just as we are able to derive the basic properties of the GCD through defining the GCD algebraically, we may also define the LCM algebraically (that is, the positive integer m such that $\mathbb{Z}m = \mathbb{Z}a \cap \mathbb{Z}b$, where m is the LCM of a and b) and develop its basic properties from there.

Proposition 1.6. Suppose $a, b \in \mathbb{Z}$, where a and b are not both zero, and let m be the least common multiple (LCM) of a and b ; that is, the positive integer m such that $\mathbb{Z}m = \mathbb{Z}a \cap \mathbb{Z}b$. It must follow that:

- i. Both a and b divide m , and
- ii. If a and b divide some number n , then m divides n as well.

Proof. Since $m \in \mathbb{Z}m$, it is contained in $\mathbb{Z}a \cap \mathbb{Z}b$, so it is contained in both $\mathbb{Z}a$ and $\mathbb{Z}b$. It follows naturally that $m = ka$ for some $k \in \mathbb{Z}$; likewise for m and b . Therefore, both a and b divide m . Furthermore, suppose n is divisible by a and b . Then $n \in \mathbb{Z}a \cap \mathbb{Z}b$, which means $n \in \mathbb{Z}m$, so m divides n , as n may be expressed as pm for some $p \in \mathbb{Z}$. $\square$

Problem 1.8

Prove that $\mathbb{Z}a \cap \mathbb{Z}b$ is a subgroup of $\mathbb{Z}$.

One important type of group is the **cyclic group**; that is, a group that is generated by an element. Cyclic groups may be finite or infinite. An example of an infinite cyclic group would be $\mathbb{Z}$, generated by the number 1, as every number can be written in the form $k1 = 1 + 1 + \cdots + 1$, where $k \in \mathbb{Z}$.

Definition 1.8 — Cyclic Group

We define a **cyclic group** $\langle x \rangle$ to be the group $\{1, x, x^2, \dots, x^{n-1}\}$, where $x^n = 1$. Note that this group has order n . Furthermore, a **cyclic subgroup** H (generated by an element x) of a group G is the set of all elements $g \in G$ such that $g = x^n$ for some $n \in \mathbb{Z}$.

Considering this previous definition, as well as the definition of the order of a group, we naturally may develop a definition of the order of an element like so:

Definition 1.9 — Order (of an element)

The **order** of an element x is the smallest positive integer n such that $x^n = 1$, the identity.

Using what we know about the integers and some of their properties, we may thus begin to prove some properties of cyclic groups themselves.

Proposition 1.7. Suppose an element $x \in G$ has a finite order n , and $k \in \mathbb{Z}$ such that $k = nq + r$, where $q, r \in \mathbb{Z}$ and $0 \leq r < n$. It must follow that

- i. $x^k = x^r$,
- ii. $x^k = 1$ if and only if $r = 0$, and
- iii. The order of x^k is equal to n/d , where d is the GCD of k and n .

Proof. We begin by showing the first claim. We have that

$$x^k = x^{nq+r} = x^{nq} \circ x^r = (x^n)^q \circ x^r = 1^q \circ x^r = 1 \circ x^r = x^r \quad (1.18)$$

Hence $x^k = x^r$. Furthermore, suppose $x^k = 1$. Then $x^r = 1$. Note that the only element $0 \leq r < n$ such that $x^r = 1$ is $r = 0$, as n is the smallest positive integer such that $x^n = 1$. In the other direction, suppose $r = 0$. Then

$$x^k = x^{nq+0} = x^{nq} = (x^n)^q = 1^q = 1 \quad (1.19)$$

It will be left to the reader to verify the third property. $\square$

Problem 1.9

Show that if x has order n , and $k \in \mathbb{Z}$, then the order of x^k is equal to n/d , where d is the GCD of k and n . *Hint:* Consider the least common multiple m of n and k . What is x^m , and why might m be important? Also, note that $nk = md$.

To continue developing our intuition about groups, we supply two more examples of groups. It is a little less obvious that these are groups. It will be left to the reader to determine whether they are cyclic groups or not.

Example: The Klein Four Group

The **Klein four group**, known as $\mathfrak{V}$ (that is a fancy V), is the group consisting of the four different matrices

$$\begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix}$$

under the operation of matrix multiplication.

Example: The Quaternions

The **quaternion group** H consists of the eight numbers

$$H = \{\pm 1, \pm i, \pm j, \pm k\}$$

where

$$i^2 = j^2 = k^2 = ijk = -1 \quad (1.20)$$

One can think of this group as extending the notion of complex numbers, where we have a number i such that $i^2 = -1$. The above equation relating quaternions was famously inscribed into Brougham Bridge by the great mathematician William Rowan Hamilton.

Problem 1.10

Is the Klein four group cyclic? Is the quaternion group cyclic?

1.3 Homomorphisms

We are concerned with the sorts of mappings between groups that preserve group operations; we call these homomorphisms. When we say “preserving group operations,” what we mean is the following definition below:

Definition 1.10 — Homomorphism

A **homomorphism** is a mapping $\varphi : G \rightarrow G'$ from a group G to a group G' such that, for all $a, b \in G$, we must have that

$$\varphi(a \circ b) = \varphi(a) \circ \varphi(b) \quad (1.21)$$

Many examples of homomorphisms are already likely familiar to you, for linear maps are homomorphisms. More information is given in the example below.

Example: Linear Maps

All **linear maps**, or maps φ between vector spaces V and W over the same field $\mathbb{F}$ such that for all $v, v' \in V$ and $a \in \mathbb{F}$,

$$\varphi(v + v') = \varphi(v) + \varphi(v') \quad (1.22)$$

and

$$\varphi(av) = a\varphi(v) \quad (1.23)$$

are examples of homomorphisms. This can be seen from the first equation. Note that vector spaces are abelian groups over the operation of vector addition.

There are certain mathematical sets that arise naturally from the concept of a homomorphism. One of these is known as the **image**, and consists of all the elements in the **codomain** (the set that is being mapped to) such that some member of the domain is mapped to such an element.

Definition 1.11 — Image

We define the **image** $\text{im } \varphi$ of a homomorphism $\varphi : G \rightarrow G'$ to be the set of all $x \in G'$ such that $x = \varphi(a)$ for some $a \in G$.

Another set that is of particular importance is the kernel, a subset of the domain group, which is the set of all elements in the group that map to the identity of the codomain group.

Definition 1.12 — Kernel

The **kernel** $\ker \varphi$ of a homomorphism $\varphi : G \rightarrow G'$ is the set of all $x \in G$ such that $\varphi(x) = 1_{G'}$.

Note that both the kernel and image of a homomorphism are subgroups of their respective groups.

Problem 1.11

Prove the above claim by considering all necessary properties of subgroups.

We denote $1_{G'}$ for the identity of the codomain group. The subscript G' demonstrates that this particular identity is the identity of the group G' .

Problem 1.12

Consider the homomorphism $\varphi : \mathbb{R}^3 \rightarrow \mathbb{R}$ given by $(x, y, z) \mapsto x$. Find the kernel of φ . Note that $\mathbb{R}$ and $\mathbb{R}^3$ are groups under the operation of addition as defined normally for both groups.

We may also be concerned with what is known as the **conjugate** of a given element, or perhaps two elements which are considered to be conjugate. The definition is a little abstract, but we present it below. Its importance will become apparent as we continue.

Definition 1.13 — Conjugate

If we have elements $a, g \in G$, where G is a group, the **conjugate** of a by g is the element $g \circ a \circ g^{-1} \in G$. Furthermore, two elements x and x' in G are considered to be **conjugate** if $x' = g \circ x \circ g^{-1}$ for some $g \in G$.

The concept of conjugacy naturally allows us to define what is known as a normal subgroup of a group, which is a particular subgroup such that any conjugate elements are also in the subgroup. The formal definition is presented below.

Definition 1.14 — Normal Subgroup

A **normal subgroup** N of a group G is a subgroup such that for every $a \in N$ and every $g \in G$, we must have that $g \circ a \circ g^{-1} \in N$.

As it turns out, the kernel of any homomorphism enjoys the properties of a normal subgroup. There are other normal subgroups that one can find, but the kernel is one of the more obvious normal subgroups.

Proposition 1.8. *The kernel of a homomorphism is a normal subgroup.*

Proof. Suppose that $a \in \ker \varphi$, where $\varphi : G \rightarrow G'$ is a homomorphism, and $g \in G$. We must have that, by the properties of a homomorphism,

$$\varphi(gag^{-1}) = \varphi(g)\varphi(a)\varphi(g^{-1}) \quad (1.24)$$

$$= \varphi(g)1_{G'}\varphi(g)^{-1} \quad (1.25)$$

$$= \varphi(g)\varphi(g)^{-1} = 1_{G'} \quad (1.26)$$

Hence, it follows that $gag^{-1} \in \ker \varphi$ as well, and therefore, the kernel of φ is a normal subgroup of G . $\square$

We have that the **center** Z of a group G is the set of all elements that *commute* with every element of G ; that is, for any $z \in Z$ and $g \in G$, we have that

$$gz = zg \quad (1.27)$$

It can be shown that the center is always a normal subgroup of G .

Problem 1.13

Show that the center of a group is always a normal subgroup of that group.

There is a special kind of homomorphism known as an isomorphism. It is essentially a homomorphism that has an inverse as well; we call such a mapping **bijective**.

Definition 1.15 — Isomorphism

An **isomorphism** $\varphi : G \rightarrow G'$ is a group homomorphism that is also bijective. Furthermore, the groups G and G' , in this case, are said to be **isomorphic**, as there exists an isomorphism between the two groups.

Problem 1.14

Show that $\mathbb{Z}$ and $2\mathbb{Z}$ are isomorphic.

Still, there is a kind of isomorphism that is also of interest; it is known as an automorphism. An isomorphism can map between two different groups; an automorphism is an isomorphism that maps a group to itself.

Definition 1.16 — Automorphism

An **automorphism** is an isomorphism $\varphi : G \rightarrow G$; it is a mapping from a group to itself.

We discuss more about automorphisms in the example below.

Example: Automorphisms

The obvious automorphism $\varphi : G \rightarrow G$ is the identity mapping, which maps $g \mapsto g$ for all $g \in G$. This is clearly an automorphism, as for every $g, g' \in G$,

$$\varphi(g \circ g') = g \circ g' = \varphi(g) \circ \varphi(g') \quad (1.28)$$

Furthermore, φ has an inverse—itsself. $\varphi \circ \varphi = \text{id}$. Finally, since φ maps G to itself, the identity mapping φ has been shown to be an automorphism. For certain groups, one can think of less trivial automorphisms. For example, we have $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ given by $x \mapsto 2x$. One can verify that this is an automorphism as well.

All automorphisms are also **endomorphisms**, which are homomorphisms from a group to itself. However, not all endomorphisms are automorphisms.

Problem 1.15

Can you think of an endomorphism that is not an isomorphism?

Just like how we can discuss a conjugate of an element with another element, it is possible to discuss what is known as the commutator of two different elements in a group. We define the commutator below:

Definition 1.17 — Commutator

The **commutator** of two elements $a, b \in G$ is the element $aba^{-1}b^{-1}$.

One final important proposition of note is the following: if we have an isomorphism, the inverse

of the isomorphism is also an isomorphism. This seems obvious *prima facie*, but we must prove it rigorously.

Proposition 1.9. *If $\varphi : G \rightarrow G'$ is an isomorphism, then it must follow that the inverse map $\varphi^{-1} : G' \rightarrow G$ is also an isomorphism.*

Proof. We already know that the inverse of a bijective map is also bijective, so φ^{-1} is bijective. Now, all we need to do is prove that φ^{-1} is a homomorphism; that is, $\varphi^{-1}(x \circ y) = \varphi^{-1}(x) \circ \varphi^{-1}(y)$. Consider the following:

$$\varphi(\varphi^{-1}(x) \circ \varphi^{-1}(y)) = \varphi(\varphi^{-1}(x)) \circ \varphi(\varphi^{-1}(y)) = x \circ y = \varphi(\varphi^{-1}(x \circ y)) \quad (1.29)$$

Since φ is bijective, as given, it naturally follows that $\varphi^{-1}(x) \circ \varphi^{-1}(y) = \varphi^{-1}(x \circ y)$. Therefore φ^{-1} is a bijective homomorphism, or an isomorphism. $\square$

1.4 Equivalence Relations

The reader may already be familiar with the concept of an equivalence relation. One example of an equivalence relation familiar from high school mathematics is that of the concept of **congruence** in geometry. Congruence enjoys several different properties; for example, a shape must always be congruent to itself (this is known as reflexivity). Furthermore, if a shape is congruent to another shape, then the other shape is congruent to the first shape (symmetry). Finally, congruence is transitive: if a shape is congruent to another shape, and that shape is congruent to a third shape, then the first shape is congruent to the third shape. We shall see that these three properties precisely define what is known as an equivalence relation; we generalize the concept from equality and congruence in geometry to encompass many different relations.

Definition 1.18 — Equivalence Relation

An **equivalence relation** is a relation $a \sim b$ between two elements a and b in a set S such that the following three properties hold:

- i. **Reflexivity:** $a \sim a$,
- ii. **Symmetry:** $a \sim b$ implies $b \sim a$, and
- iii. **Transitivity:** $a \sim b$ and $b \sim c$ imply $a \sim c$.

We now know what an equivalence relation is, but what precisely is a **relation**? One can define it as follows: a relation R of a set S is a subset of the Cartesian product of S with itself; that is, $R \subset S \times S$. An equivalence relation is simply a relation R such that, given $a, b, c \in S$, we have that

1. $(a, a) \in R$ (reflexivity),
2. $(a, b) \in R \implies (b, a) \in R$ (symmetry), and
3. $(a, b), (b, c) \in R \implies (a, c) \in R$ (transitivity).

Thinking about relations in this way is a highly unusual and abstract way to do so, so it may also be helpful to think of relations another way. A relation describes a relationship between two objects in a given set S . Suppose we have an object a . a might not be related to any objects in S . a could be related to one object in S , or perhaps even more than one object in S . Furthermore, unlike equivalence relations, relations are not necessarily symmetric; it is possible for a to be related to b and not the other way around.

Shifting our focus back to equivalence relations, it is possible to develop something known as an equivalence class from them. An equivalence class of an element $a \in S$ is essentially the set of all elements that are equivalent to a based on the equivalence relation $\sim$.

Definition 1.19 — Equivalence Class

The **equivalence class** of some element a in a set S , denoted by $[a]$, is defined to be the set

$$[a] = \{b \in S \mid a \sim b\} \quad (1.30)$$

We may also find it useful to consider the concept of partitions of a set. This is exactly what it sounds like; the set is partitioned into different subsets such that all elements in the set are in only one subset and none of the subsets are empty.

Definition 1.20 — Partition

We define a **partition** Π of a set S to be a subdivision of S into subsets P_α such that $P_\alpha \neq \emptyset$ for all α , the union of all P_α is S , and, if $a \in P_i$, then $a \notin P_\alpha$ for any other α .

We will examine a very simple example of a partition of a set below.

Example: Odd and Even Numbers

Consider the set of integers $\mathbb{Z}$. We can partition the set $\mathbb{Z}$ into **odd** and **even** numbers:

$$O = \{n \in \mathbb{Z} \mid n = 2k + 1, \text{ where } k \in \mathbb{Z}\} \quad (1.31)$$

$$E = \{n \in \mathbb{Z} \mid n = 2k, \text{ where } k \in \mathbb{Z}\} \quad (1.32)$$

One can see that $O \neq \emptyset$ and $E \neq \emptyset$, as $1 \in O$ and $0 \in E$, for example. Furthermore, $O \cup E = \mathbb{Z}$, as every integer is either odd or even. Finally, we have that the sets are disjoint; there is no number that is both odd and even at the same time. Hence, we have partitioned the set $\mathbb{Z}$.

It is now time to prove a lemma regarding equivalence relations and partitions. As it turns out, the two concepts are rather connected. This lemma will serve to help prove a theorem later in the chapter.

Lemma 1.10. *If we have an equivalence relation $\sim$ on a set S , the subsets of S that are equivalence classes must partition S .*

Proof. We must show two different properties. We must show that the equivalence classes of $\sim$ are all nonempty, we must show that the union of the equivalence classes is the entire set, and we must show that they are disjoint. To show that they are nonempty, all we need to do is consider the reflexive property of the equivalence relation: $a \sim a$ for all $a \in S$, so $a \in [a]$. Therefore, no matter what equivalence class we choose, the subset will be nonempty. Furthermore, since every element belongs to its own equivalence class, we have that the union of the equivalence classes is the set S itself. Now all that remains is to show that the equivalence classes are disjoint; that is, no element belongs to two or more equivalence classes. This will be left to the reader. $\square$

Problem 1.16

Show that the equivalence classes are disjoint. Do this by assuming that $[a]$ and $[b]$ have an element in common. Then show that $[a] = [b]$ by proving that $[a] \subset [b]$, and $[b] \subset [a]$.

Note that if we have a mapping $f : S \rightarrow T$, we are able to define an equivalence relation based on this map. We do so as follows: given $a, b \in S$, we state that $a \sim b$ if $f(a) = f(b)$, where equality is defined regularly on T . To show that this relation is reflexive, we consider that $f(a) = f(a)$, so $a \sim a$. To show that this relation is symmetric, assuming that $a \sim b$, then we have that $f(a) = f(b)$, so obviously $f(b) = f(a)$, and therefore $b \sim a$. To show that the relation is transitive, assuming that $a \sim b$ and $b \sim c$, we have that $f(a) = f(b)$ and $f(b) = f(c)$, so it naturally follows that $f(a) = f(c)$, so $a \sim c$. It is thus clear that the defined relation $\sim$ is indeed an equivalence relation.

The concept of a fiber is particularly useful in geometry and topology when considering objects known as **fiber bundles**, but these mathematical objects will not be covered in this book; they require some knowledge of differential geometry in order to deeply understand. However, we will discuss one aspect of fiber bundles, which is, of course, the fiber. It is also known as the inverse image, which is slightly more telling regarding what it actually means.

Definition 1.21 — Fiber

A **fiber**, also known as an **inverse image**, of an element $t \in T$, given a mapping $f : S \rightarrow T$, is the subset of S containing all elements s where $f(s) = t$.

To understand what fibers are precisely, an example is provided down below that should serve to highlight the concept of a fiber in a nontrivial way.

Example: Fibers

Consider the map $f : \mathbb{Z} \rightarrow \mathbb{Z}_3$, where $\mathbb{Z}_3 = \{0, 1, 2\}$ equipped with the operation of modular addition (it is not important to know what that is for now). We define this map as follows: $f(x) = x \bmod 3$, where $x \bmod 3$ is the remainder when x is divided by 3. The fiber $f^{-1}(0)$ consists of all elements x such that their remainder, when divided by 3, is 0; this is just all multiples of 3, or $3\mathbb{Z}$. Similarly, we are able to discern the fibers $f^{-1}(1)$ and $f^{-1}(2)$.

It is also worth noting the concept of congruence, as defined in abstract algebra. This is a specific type of equivalence relation naturally inherited from a mapping that is also a homomorphism, as we can see below.

Definition 1.22 — Congruence

We say that an equivalence relation is called **congruence** if the equivalence relation is defined by a group homomorphism $\varphi : G \rightarrow G'$, like so:

$$a \equiv b \quad \text{if} \quad \varphi(a) = \varphi(b) \quad (1.33)$$

Problem 1.17

Consider the equivalence relation naturally admitted from the mapping $\varphi : \mathbb{R}^3 \rightarrow \mathbb{R}$ where $(x, y, z) \mapsto x + y$. What is the equivalence class $[(16, 23, 30)]$? What form do fibers take, algebraically and geometrically? Is this equivalence relation a congruence?

1.5 Cosets

One interesting mathematical object that we may consider is that of the coset, which is simply the set of the elements of a subgroup composed with an element in the group. More precisely, we define a left coset as follows:

Definition 1.23 — Left Coset

A **left coset** aH of a subgroup H of a group G is defined as

$$aH = \{g \in G \mid g = ah \text{ for some } h \in H\} \quad (1.34)$$

Problem 1.18

Is aH always a group? Why?

It is worthwhile to mention that there exist right cosets as well, which are exactly what you might think they are, but we will consider mainly left cosets here.

As it turns out, if we have a homomorphism, the cosets of the kernel partition the domain group, and each coset is associated with an element in the image, as we can prove below.

Theorem 1.11. *Suppose we have $K = \ker \varphi$ of a homomorphism $\varphi : G \rightarrow G'$. We must have that the fiber of φ containing an element $a \in G$ is the coset aK of the kernel. They partition the group G , and furthermore are each associated with individual elements of the image.*

Proof. We begin by showing that the fiber F of φ containing an element $a \in G$ is equivalent to the coset aK . We do so by showing that $F \subset aK$ and $aK \subset F$. Suppose we have an element $f \in F$. It follows that $\varphi(f) = \varphi(a)$, as F contains a . Thus, we have that

$$\varphi(a^{-1} \circ f) = \varphi(a^{-1}) \circ \varphi(f) = \varphi(a)^{-1} \circ \varphi(a) = 1_{G'} \quad (1.35)$$

Therefore, $a^{-1} \circ f \in K$, so it naturally follows that $f = a \circ (a^{-1} \circ f) \in aK$. Therefore, $F \subset aK$. In the other direction, suppose we have some $b = a \circ k \in aK$. It follows that

$$\varphi(b) = \varphi(a \circ k) = \varphi(a) \circ \varphi(k) = \varphi(a) \circ 1_{G'} = \varphi(a) \quad (1.36)$$

Hence, we must have that $b \in F$, so $aK \subset F$, and therefore $F = aK$. So, the fiber of φ containing $a \in G$ must be the coset aK of the kernel. Since a and b are in the same fiber if $\varphi(a) = \varphi(b)$, this naturally admits an equivalence relation based on the mapping φ (we have already shown that such a relation is an equivalence relation). In this case, we have that $[a]$ is the fiber containing $a \in G$. By Lemma 1.10, which states that the equivalence classes of an equivalence relation partition a

set, these equivalence classes must therefore partition G , and furthermore, since $\varphi(a) = \varphi(b)$ for any $b \in [a]$ and no other elements, it follows that each equivalence class corresponds to a distinct element in the image in G' . $\square$

Problem 1.19

Show that the cosets of a subgroup always partition the group. Examine the proof of Theorem 1.11.

Knowing that the cosets of a subgroup always partition the group, it is naturally an important affair of discerning how many possible cosets of a subgroup there are. We have a term for this, known as the index:

Definition 1.24 — Index

The **index** of a subgroup H in G (denoted $[G : H]$) is the number of left cosets of H .

We can determine a bijective correspondence between any subgroup H of a group G and its coset aH , where $a \in G$. Consider the mapping $f : H \rightarrow aH$, where $h \mapsto ah$. This map has a defined inverse $f^{-1} : aH \rightarrow H$, where $ah \mapsto a^{-1}ah = h$. Therefore, we conclude the following.

Remark

We see that all left cosets aH of a subgroup H of a group G must have the same order.

This allows us to prove the incredibly useful **Counting Formula**, which allows us to express the order of G in terms of the order of a subgroup H and its index.

Theorem 1.12 (Counting Formula). For any group G and a subgroup H , the following equation holds:

$$|G| = |H|[G : H] \quad (1.37)$$

where $|G|$ denotes the order of G .

Proof. We are aware that all cosets aH for all $a \in G$ have the same order as H . Furthermore, the cosets partition the group, which implies that all elements of G belong to exactly one coset. Therefore, we see that the number of elements of G can be found by multiplying the number of elements of H (or any coset) by the number of cosets of H in G , or $[G : H]$. Hence, we have obtained the desired expression. $\square$

A direct result of the Counting Formula is the famous **Lagrange's Theorem**, named after the mathematician Joseph-Louis Lagrange. It is stated and proven below.

Theorem 1.13 (Lagrange's Theorem). If H is a subgroup of a finite group G , then the order of H divides the order of G .

Proof. If G is a finite group, then $|G|, |H|, [G : H] \in \mathbb{Z}$. Since $|G| = |H|[G : H]$, we see that $|H|$ divides $|G|$, as $|G|/|H| = [G : H]$, which is an integer. Therefore, the order of H divides the order of G . $\square$

A useful corollary of the Counting Formula deals with the kernel and image of a homomorphism. As it turns out, they are related in the following way.

Corollary 1.14. *Suppose that $\varphi : G \rightarrow G'$ is a homomorphism, and G, G' are finite groups. We must therefore have that*

$$|G| = |\ker \varphi| \cdot |\operatorname{im} \varphi| \quad (1.38)$$

Proof. We have already established before that $\ker \varphi$ is a subgroup of G . Therefore, by the Counting Formula, we must have that

$$|G| = |\ker \varphi| \cdot [G : \ker \varphi] \quad (1.39)$$

However, recall by Theorem 1.11 that the number of cosets of $\ker \varphi$ is the order of the image, as each coset is associated with one element of the image. Therefore, we have that

$$|G| = |\ker \varphi| \cdot |\operatorname{im} \varphi| \quad (1.40)$$

which is our desired expression. $\square$

Problem 1.20

Show that $|\ker \varphi|$ divides $|G|$, and $|\operatorname{im} \varphi|$ divides both $|G|$ and $|G'|$. *Hint:* Consider the result of Problem 1.11.

At last, we come to the ultimate result of this section. It is known as the **Correspondence Theorem**, and essentially deals with the fact that if we have a surjective homomorphism, there is a bijective correspondence between subgroups of the domain containing the kernel and subgroups of the image. The correspondence is defined explicitly in the theorem. There are some other useful consequences that are written in the theorem and proven as well.

Theorem 1.15 (The Correspondence Theorem). *Suppose $\varphi : G \rightarrow \mathcal{G}$ is a surjective homomorphism. There exists a bijection between subgroups of $\mathcal{G}$ and subgroups of G containing the kernel K , where a subgroup H of G containing K is associated with its image $\varphi(H)$, and a subgroup $\mathcal{H}$ of $\mathcal{G}$ is associated with its inverse image (note that it is not a fiber, as $\mathcal{H}$ may contain more than one element) $\varphi^{-1}(\mathcal{H})$. Furthermore, if H and $\mathcal{H}$ correspond as above, then H is a normal subgroup of G if and only if $\mathcal{H}$ is a normal subgroup of $\mathcal{G}$. Finally, $|H| = |\mathcal{H}| \cdot |K|$.*

Proof. We have a lot to prove here; five things, in fact.

First, we prove that $\varphi(H)$ really is a subgroup of $\mathcal{G}$. Consider the restriction of the homomorphism φ to H , which we denote by $\varphi|_H : H \rightarrow \mathcal{G}$. The image of H , $\varphi(H)$, is therefore a group, and is hence a subgroup of $\mathcal{G}$.

Now, we must prove that $\varphi^{-1}(\mathcal{H})$ is a subgroup of G which contains K . Suppose that we have any $k \in K$. Then $\varphi(k) = 1_{\mathcal{G}}$, which is in $\mathcal{H}$, as $\mathcal{H}$ is a subgroup of $\mathcal{G}$. Hence, $K \subset \varphi^{-1}(\mathcal{H})$, and so the inverse image of $\mathcal{H}$ contains K . Now we must show that $\varphi^{-1}(\mathcal{H}) = H$ is a subgroup. If $g, h \in H$, then $\varphi(g), \varphi(h) \in \mathcal{H}$. $\mathcal{H}$ is a subgroup, so $\varphi(g) \circ \varphi(h) \in \mathcal{H}$. Recall that φ is a homomorphism, so $\varphi(g) \circ \varphi(h) = \varphi(g \circ h) \in \mathcal{H}$, so $g \circ h \in H$. Therefore, H satisfies the closure property. Furthermore, $\varphi(1_G) = 1_{\mathcal{G}} \in \mathcal{H}$, so $1_G \in H$, and so H satisfies the identity property.

Finally, if $g \in H$, then $\varphi(g) \in \mathcal{H}$. As $\mathcal{H}$ is a subgroup, we have that $\varphi(g)^{-1} \in \mathcal{H}$, and since φ is a homomorphism, $\varphi(g^{-1}) \in \mathcal{H}$, so $g^{-1} \in H$. Hence, H satisfies the inverse property. Therefore, $\varphi^{-1}(\mathcal{H})$ is indeed a subgroup of G which contains K .

We have established that this correspondence really is between subgroups of G containing K and subgroups of $\mathcal{G}$. Now we must show that this correspondence is bijective. Let H and $\mathcal{H}$ be corresponding subgroups of their respective groups, with H containing K . It is clear that $\varphi(\varphi^{-1}(\mathcal{H})) = \mathcal{H}$, as φ is given to be surjective. Furthermore, we know that $H \subset \varphi^{-1}(\varphi(H))$, as if $h \in H$, then $\varphi(h) \in \varphi(H)$, so $h \in \varphi^{-1}(\varphi(H))$. However, this is not good enough. To prove the correspondence is bijective, we must also show that $\varphi^{-1}(\varphi(H)) = H$, which means we only need to show that $\varphi^{-1}(\varphi(H)) \subset H$. Suppose $a \in \varphi^{-1}(\varphi(H))$. It follows that $\varphi(a) \in \varphi(H)$. There exists some $h \in H$ such that $\varphi(a) = \varphi(h)$. Therefore, $\varphi(h)^{-1} \circ \varphi(a) = 1_G$, so $\varphi(h^{-1}) \circ \varphi(a) = 1_G$, and $\varphi(h^{-1} \circ a) = 1_G$, and so $h^{-1} \circ a \in K$. Since $K \subset H$, we have that $h^{-1} \circ a \in H$, and as $h \in H$ as well, it naturally follows that $h \circ (h^{-1} \circ a) = a \in H$. Therefore, $a \in \varphi^{-1}(\varphi(H))$ implies $a \in H$, so $\varphi^{-1}(\varphi(H)) \subset H$, and so $\varphi^{-1}(\varphi(H)) = H$. All this ultimately shows that our correspondence is bijective, as $\varphi(\varphi^{-1}(\mathcal{H})) = \mathcal{H}$ and $\varphi^{-1}(\varphi(H)) = H$.

With this in mind, we may now show that $\mathcal{H}$ is a normal subgroup of $\mathcal{G}$ if and only if $\varphi^{-1}(\mathcal{H})$ is a normal subgroup of G . We will prove one direction, and leave the other direction for the reader. Suppose $\mathcal{H}$ is normal. Suppose we choose an arbitrary $h \in \varphi^{-1}(\mathcal{H})$ and $g \in G$. We therefore have that $\varphi(g \circ h \circ g^{-1}) = \varphi(g) \circ \varphi(h) \circ \varphi(g)^{-1}$, which is a conjugate of $\varphi(h)$, which is also in $\mathcal{H}$, as $\mathcal{H}$ is a normal subgroup. Therefore, $g \circ h \circ g^{-1} \in \varphi^{-1}(\mathcal{H})$, so $\varphi^{-1}(\mathcal{H})$ is a normal subgroup of G . The other direction is, as stated before, left for the reader to prove.

Finally, we must show that $|H| = |\mathcal{H}| \cdot |K|$. It is known that $K \subset H$, and both K and H are subgroups and finite. Consider the homomorphism $\varphi|_H : H \rightarrow \mathcal{G}$. We have that, by Corollary 1.14,

$$|H| = \left| \ker \varphi|_H \right| \cdot \left| \text{im } \varphi|_H \right| = K \cdot \left| \text{im } \varphi|_H \right| \quad (1.41)$$

But $\left| \text{im } \varphi|_H \right|$ is precisely $|\mathcal{H}|$ by our defined correspondence, so

$$|H| = |\mathcal{H}| \cdot |K| \quad (1.42)$$

as desired.

We have shown all five claims of the Correspondence Theorem that required proof. $\square$

Problem 1.21

Show that, given the conditions defined in the Correspondence Theorem (specifically, that φ is a surjective homomorphism), if H is a normal subgroup of G , then $\varphi(H) = \mathcal{H}$ is a normal subgroup of $\mathcal{G}$.

1.6 Modular Arithmetic

We say that a and b are **congruent modulo** n , or $a \equiv b \pmod{n}$, if n divides the number $b - a$. One can see that this congruence fulfills the properties of an equivalence relation. Therefore, we may

consider what we call the **congruence classes**, or equivalence classes, defined by this relation. We define:

$$\bar{a} = \{\dots, a - n, a, a + n, \dots\} \quad (1.43)$$

One can see that

$$\bar{0} = \mathbb{Z}n = \{kn \mid k \in \mathbb{Z}\} \quad (1.44)$$

Furthermore, the other congruence classes are simply cosets of $\bar{0}$ with respect to the operation of addition—note that the standard notation aH would not be appropriate here, so instead we shall use the notation

$$\bar{a} = a + \bar{0} = \{a + kn \mid k \in \mathbb{Z}\} \quad (1.45)$$

Note that there are n integers from 0 to $n - 1$.

We shall show that the congruence classes of these n integers are the only congruence classes modulo n .

Proposition 1.16. *There are exactly n congruence classes modulo n , and so $[\mathbb{Z} : \mathbb{Z}n] = n$.*

Proof. Suppose that there exists any integer p . Note that we can say that $p = nq + r$ for some $q, r \in \mathbb{Z}$, and $0 \leq r < n$. Note that $p - r = nq$, which is divisible by n , so p is in the congruence class of r , but r can take any value between and including 0 and $n - 1$. There are n integers from 0 to $n - 1$. Hence, there exist n congruence classes. Since these congruence classes are cosets of $\mathbb{Z}n$, it follows that the number of cosets of $\mathbb{Z}n$, or $[\mathbb{Z} : \mathbb{Z}n]$, is equal to n . $\square$

As it turns out, we are able to add and multiply congruence classes; that is, we are able to speak of $\bar{a} + \bar{b} = \overline{a + b}$ and $\bar{a}\bar{b} = \overline{ab}$. The theorem below shows that this relation is true.

Theorem 1.17 (Modular Arithmetic). *Suppose $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$. It follows that $a' + b' \equiv a + b \pmod{n}$ and $a'b' \equiv ab \pmod{n}$.*

Proof. We show that $a' + b' \equiv a + b \pmod{n}$. Let $a' = a + rn$ and $b' = b + sn$ for some $r, s \in \mathbb{Z}$. Then we must have

$$a' + b' = a + rn + b + sn = a + b + (r + s)n \quad (1.46)$$

$$\implies (a' + b') - (a + b) = (r + s)n \quad (1.47)$$

This quantity is clearly divisible by n , so $a' + b' \equiv a + b \pmod{n}$. Now, we show that $a'b' \equiv ab \pmod{n}$. We have

$$a'b' = (a + rn)(b + sn) = ab + as n + br n + rs n^2 \quad (1.48)$$

$$= ab + (as + br + rs n)n \quad (1.49)$$

$$\implies a'b' - ab = (as + br + rs n)n \quad (1.50)$$

This quantity is also divisible by n , so $a'b' \equiv ab \pmod{n}$. $\square$

Example: Using Modular Arithmetic

Suppose that we want to find the last digit of 3^{67} . One might realize that this is equivalent to figuring out $3^{67} \pmod{10}$. We know that $3 \pmod{10} = 3$, and $3^2 \pmod{10} = 9 \pmod{10} = 9$. Furthermore, we have that $3^3 \pmod{10} = 9 \cdot 3 \pmod{10} = 27 \pmod{10} = 7$. Then, we use

the property that $\overline{a}\overline{b} = \overline{ab}$ to find $3^4 \bmod 10 = (3^3 \cdot 3) \bmod 10 = 3^3 \bmod 10 \cdot 3 \bmod 10 = 7 \cdot 3 \bmod 10 = 21 \bmod 10 = 1$. It necessarily follows that $3^5 \bmod 10 = 3$, and so on. One can see that if we have some k divisible by four, then $3^k \bmod 10 = 3^{4n} \bmod 10$ for some $n \in \mathbb{Z}$, which is equal to $(3^4)^n \bmod 10 = 1^n \bmod 10 = 1$. We then say

$$3^{67} = 3^{64} \cdot 3^3 \bmod 10 = 1 \cdot 3^3 \bmod 10 = 7 \quad (1.51)$$

The last digit in 3^{67} is 7.

We denote the set of congruence classes modulo n by the symbol $\mathbb{Z}/n\mathbb{Z}$. It will become apparent soon enough why we choose to write it as a *quotient* of sorts.

Problem 1.22

(Source: 1992 AHSME #17) Suppose we construct the integer

$$N = 192021 \cdots 909192 \quad (1.52)$$

by writing out the integers from 19 to 92 in consecutive order. Let 3^k be the largest power of 3 that divides N . Find k .

1.7 Product and Quotient Groups

We now consider product groups. A product group of two groups is essentially the Cartesian product of two groups equipped with a law of composition defined below.

Definition 1.25 — Product Group

We say that the **product group** $G \times G'$ of two groups G and G' is the set of ordered pairs (g, g') , where $g \in G$ and $g' \in G'$, equipped with the law of composition defined as follows:

$$(g, g') \circ (h, h') = (g \circ h, g' \circ h') \quad (1.53)$$

We now turn our attention to the multiplication map $f : H \times K \rightarrow G$, where H and K are subgroups of G , and $(h, k) \mapsto hk$. This map may enjoy special properties under certain conditions, outlined and proven below.

Proposition 1.18. Suppose H and K are subgroups of a group G , with a mapping $f : H \times K \rightarrow G$ defined by sending $(h, k) \mapsto hk$. The image of f is denoted $HK = \{hk \mid h \in H, k \in K\}$. We have that:

- i. f is injective if and only if $H \cap K = \{1\}$,
- ii. f is a homomorphism from $H \times K$ to G if and only if $hk = kh$, where $h \in H$ and $k \in K$,
- iii. If H is a normal subgroup of G , then HK is a subgroup of G , and
- iv. f is an isomorphism from $H \times K$ to G if and only if $H \cap K = \{1\}$, $HK = G$, and both H and K are normal subgroups of G .

Proof. We have four things to prove here. Let's get to work.

We show the contrapositive of the statement: if f is injective, then $H \cap K = \{1\}$. That is, we will show that if $H \cap K \neq \{1\}$, then f is not injective. Suppose there is an element $x \in H \cap K$ that is not the identity. Then $x^{-1} \in H$, and $x \in K$ as well, so $f(x^{-1}, x) = x^{-1}x = 1$, but $f(1, 1) = 1$ as well, as $1 \in H \cap K$. Therefore, f is not injective. In the other direction, we shall show that if $H \cap K = \{1\}$, then f is injective. Suppose that $(h_1, k_1), (h_2, k_2) \in H \times K$, and $h_1k_1 = h_2k_2$. We have

$$h_1^{-1}(h_1k_1)k_2^{-1} = h_1^{-1}(h_2k_2)k_2^{-1} \implies k_1k_2^{-1} = h_1^{-1}h_2 \quad (1.54)$$

Note that $k_1k_2^{-1} \in K$ and $h_1^{-1}h_2 \in H$, so both of these expressions, being in $H \cap K$, must be 1. Therefore, $k_1k_2^{-1} = h_1^{-1}h_2 = 1$, and so $k_1 = k_2$, and $h_1 = h_2$, so $(h_1, k_1) = (h_2, k_2)$, so f is injective. Hence, f is injective if and only if $H \cap K = \{1\}$.

Suppose that f is a homomorphism; that is, suppose $f(h_1h_2, k_1k_2) = f(h_1, k_1)f(h_2, k_2)$. We see that $f(h_1h_2, k_1k_2) = h_1h_2k_1k_2$ and $f(h_1, k_1)f(h_2, k_2) = h_1k_1h_2k_2$. Therefore, we must have that

$$h_1h_2k_1k_2 = h_1k_1h_2k_2 \implies h_2k_1 = k_1h_2 \quad (1.55)$$

for any $h_2 \in H$ and $k_1 \in K$. To prove the backwards claim, we can simply go backwards in our line of logic. Therefore, f is a homomorphism from $H \times K$ to G if and only if $hk = kh$ for all $h \in H$ and $k \in K$.

Now, suppose that H is a normal subgroup of G . We will show that HK is a subgroup of G . Consider that KH is the union of all left cosets kH , where $k \in K$, and the same thing for HK . Therefore, we have that $kH = Hk$, since H is normal, as $khk^{-1} = h' \in H \implies kh = h'k$ for some $k \in K, h, h' \in H$. Therefore, the union of all kH should be equal to the union of all Hk , so $KH = HK$. It follows that HK is closed under multiplication, since

$$HKHK = H(KH)K = H(HK)K = HHKK = HK \quad (1.56)$$

Furthermore, the identity is of course in HK , as the identity is in both H and K . Finally, every element $hk \in HK$ has an inverse $(hk)^{-1} \in HK$, as $k^{-1}h^{-1} \in KH = HK$, and this expression can be shown to be an inverse of hk . Therefore, if H is a normal subgroup of G , it follows that HK is a subgroup of G .

If f is an isomorphism, then f is bijective, so f is injective, so by (i), $H \cap K = \{1\}$. Furthermore, f must also be surjective, so the image HK must be that of G , and so $HK = G$. f is a homomorphism, so $hk = kh$ for $h \in H$ and $k \in K$. Consider an arbitrary element $g \in G$, which can be represented as some $h'k'$, where $h' \in H$ and $k' \in K$, as $HK = G$. Then,

$$ghg^{-1} = h'k'hk'^{-1}h'^{-1} = h'k'k'^{-1}hh'^{-1} = h'hh'^{-1} \in H \quad (1.57)$$

so H is a normal subgroup of G . One can repeat this argument for K . Therefore, if f is an isomorphism from $H \times K$ to G , all of those conditions follow. Now, we prove this in the opposite direction. Since $H \cap K = \{1\}$, f is injective. Since $HK = G$, f is surjective. Therefore, f is bijective. We now consider the commutator $(hkh^{-1})k^{-1} = h(kh^{-1}k^{-1})$ for some $h \in H, k \in K$. K is normal, so $hkh^{-1} \in K$ and therefore the left hand side is in K . Similarly, since H is normal, the right hand side is in H . Therefore, the commutator is in both H and K , so it must be the identity. So, $hkh^{-1}k^{-1} = 1$ implies $hk = kh$. This implies that, by (ii), f is a homomorphism. Therefore, f is a bijective homomorphism, or an isomorphism.

We have thus proved all four claims. □

Problem 1.23

Show that, up to isomorphism, there are only two groups of order 4, the cyclic group C_4 of order four and the Klein four group, which is isomorphic to $C_2 \times C_2$. *Hint:* Consider the possible orders of elements in a group of order four, and use the previous proposition to show that one of the cases is isomorphic to $C_2 \times C_2$.

We now consider the quotient group. This is slightly less intuitive than the product group, but we say that G/N is the set of all cosets of N , which is a normal subgroup of G .

Definition 1.26 — Quotient Group

The **quotient group** G/N is the set of cosets of a normal subgroup N of a group G .

It is now that we finally understand the meaning behind the notation of $\mathbb{Z}/n\mathbb{Z}$, as is explained below.

Example: $\mathbb{Z}/n\mathbb{Z}$

Recall that each congruence class modulo n is a coset of $\bar{0}$. Furthermore, we have that $\bar{0}$ is a normal subgroup. To verify this, consider an element $a \in \mathbb{Z}$, with inverse $-a$. We have that

$$a + kn - a = kn \in \bar{0} \quad (1.58)$$

where $\bar{0} = \{kn \mid k \in \mathbb{Z}\}$, as discussed before. Hence, it is clear that $\bar{0}$ is a normal group. Therefore, by the definition of a quotient group, the set of congruence classes modulo n of $\mathbb{Z}$ is a quotient group, $\mathbb{Z}/\bar{0} = \mathbb{Z}/n\mathbb{Z}$.

Problem 1.24

Recall that $\mathbb{R}$ is a group under addition. Consider the normal subgroup $\mathbb{Z} \subset \mathbb{R}$ (verify that the integers are indeed such a normal subgroup of $\mathbb{R}$). If we take the quotient group $\mathbb{R}/\mathbb{Z}$, we obtain a group that shares very similar properties to a certain shape. What is that shape?

We are now able to consider what is known (in Artin's text) as the First Isomorphism Theorem. It states that if we have a homomorphism $\varphi : G \rightarrow G'$, the quotient group G/K , where $K = \ker \varphi$, is isomorphic to G' .

Theorem 1.19 (The First Isomorphism Theorem). Suppose that $\varphi : G \rightarrow G'$ is a surjective group homomorphism, with a kernel K . Define $\bar{G} = G/K$. There exists a unique isomorphism $\bar{\varphi} : \bar{G} \rightarrow G'$ such that, if we have $\pi : G \rightarrow \bar{G}$ being the canonical map (that is, the map taking g to its respective coset), then $\varphi = \bar{\varphi} \circ \pi$.

Proof. We have seen before that the cosets of K , or the elements of $\bar{G}$, are the fibers of the map φ . It is clear then that $\bar{\varphi}$ sends some nonempty fiber $\bar{g}$ to the image of the fiber, like so:

$$\bar{\varphi}(\bar{g}) = \varphi(g) \quad (1.59)$$

Note that φ is surjective. We will show that $\bar{\varphi}$ is bijective by showing that it has an inverse $\bar{\varphi}^{-1}$.

We define $\bar{\varphi}^{-1}$ to map some $g' \in G'$ to its fiber $\varphi^{-1}(g')$. Since φ is surjective, the fiber $\varphi^{-1}(g')$ is never empty. If we denote the coset aK by $\bar{a}$, and letting $\varphi(g) = g'$, it is clear to see that

$$\bar{\varphi}(\bar{\varphi}^{-1}(g')) = \bar{\varphi}(\bar{g}) = g' \quad \text{and} \quad \bar{\varphi}^{-1}(\bar{\varphi}(\bar{g})) = \bar{\varphi}^{-1}(g') = \bar{g} \quad (1.60)$$

Hence $\bar{\varphi}$ is bijective. Furthermore, since φ is a homomorphism, $\bar{\varphi}$ too is a homomorphism, as

$$\bar{\varphi}(\bar{g}\bar{h}) = \varphi(gh) = \varphi(g)\varphi(h) = \bar{\varphi}(\bar{g})\bar{\varphi}(\bar{h}) \quad (1.61)$$

Thus, $\bar{\varphi}$ is a bijective homomorphism, indeed, an isomorphism. $\square$

Problem 1.25

Show that if $\varphi : G \rightarrow G'$ is a homomorphism, with kernel K and an image H' , then G/K is isomorphic to H' .

1.8 Exercises

Exercises

1. Prove that every subgroup of a cyclic group is cyclic.
2. Suppose $\varphi : G \rightarrow G'$ is a surjective homomorphism. If G is cyclic, show that G' is cyclic. If G is abelian, show that G' is abelian.
3. Suppose H and K are subgroups of G . Show that the intersection $xH \cap yK$ of two cosets of H and K is either empty or a coset of $H \cap K$.
4. Suppose P is a partition of a group G such that for any subsets A, B of G in P , the product set AB is a subset of some $C \in P$. Let $N \in P$ contain the identity. Show that N is a normal subgroup of G and P is the set of all of its cosets.
5. Suppose that the set of automorphisms of a group G is cyclic. Is G abelian?
6. Define $(\mathbb{Z}/n\mathbb{Z})^\times$ as the set of integers relatively prime to n modulo n ; this is a group under multiplication. Prove **Euler's Theorem**: $a^{\varphi(n)} \equiv 1 \pmod{n}$ for all $a \in (\mathbb{Z}/n\mathbb{Z})^\times$, where φ is **Euler's totient function**: $\varphi(n)$ is the number of positive integers up to n that are relatively prime to n (for example $\varphi(2) = 1$, as 1 is relatively prime to 2, so there is 1 number below 2 relatively prime to 2).
7. Consider $a, b \in \mathbb{R}^k$. We define a **path** from a to b as a continuous function $\varphi : [0, 1] \rightarrow \mathbb{R}^k$ such that $\varphi(0) = a$ and $\varphi(1) = b$. Suppose $S \subset \mathbb{R}^k$, we say that $a \sim b$ if there exists a path whose image lies entirely in S from a to b .
 - i. Show that $\sim$ is an equivalence relation on S .
 - ii. A subset S is said to be **path connected** if $a \sim b$ for any $a, b \in S$. Show that any subset $S \subset \mathbb{R}^k$ can be partitioned into path-connected subsets such that two points in different subsets cannot be connected by a path in S .
 - iii. Which sets are path connected? $\{(x, y) \mid x^2 + y^2 = 1\}, \{(x, y) \mid xy = 0\}, \{(x, y) \mid xy = 1\}$

Symmetry

“What is it indeed that gives us the feeling of elegance in a solution, in a demonstration? It is the harmony of the diverse parts, their symmetry, their happy balance; in a word it is all that introduces order, all that gives unity, that permits us to see clearly and to comprehend at once both the ensemble and the details.”

Henri Poincaré

2.1 Isometries

The first definition that we will come across in this chapter is that of an isometry. An isometry is essentially a mapping that preserves distance. Consider the Euclidean norm $|\cdot|$ equipped to $\mathbb{R}^n$. Then we define an isometry like so:

Definition 2.1 — Isometry

We define an **isometry** $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ to be a mapping such that, for all $x, y \in \mathbb{R}^n$, we have that

$$|f(x) - f(y)| = |x - y| \quad (2.1)$$

There are many different mappings that may be familiar to you that are isometries. For example, orthogonal linear operators are isometries, as they preserve the dot product, and hence preserve the norm. As we can see below, translations are isometries as well.

Example: Translations

Consider the translation $t_v : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by a vector $v \in \mathbb{R}^3$, where $t_v(x) = x + v$. This map is an isometry. Suppose we have $x, y \in \mathbb{R}^3$. Then

$$|t_v(x) - t_v(y)| = |(x + v) - (y + v)| = |x + v - y - v| = |x - y| \quad (2.2)$$

Hence t_v is an isometry.

Problem 2.1

Is t_v a linear map?

Let us consider $\mathbb{R}^3$. If we have a mapping φ that is an isometry that fixes the origin, this gives rise to some interesting properties. We will show that it preserves the dot product, and also that it must be an orthogonal linear operator on $\mathbb{R}^3$.

Theorem 2.1. *Suppose we have a mapping $\varphi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. These properties are equivalent (that is, if φ has one, it has them all):*

- i. φ is an isometry, and $\varphi(0) = 0$,
- ii. $\varphi(v) \cdot \varphi(w) = v \cdot w$ for any $v, w \in \mathbb{R}^3$, and
- iii. φ is an orthogonal linear operator.

Proof. We will begin by proving (iii) implies (i), then (ii) implies (iii), and finally (i) implies (ii).

(iii) $\implies$ (i): Suppose φ is an orthogonal linear operator. Since it is a linear map, it follows that

$$\varphi(v) - \varphi(w) = \varphi(v - w) \quad (2.3)$$

so we have

$$|\varphi(v) - \varphi(w)| = |\varphi(v - w)| = (\varphi(v - w) \cdot \varphi(v - w))^{1/2} \quad (2.4)$$

Note that, since φ is orthogonal, we must have that $\varphi(v - w) \cdot \varphi(v - w) = (v - w) \cdot (v - w)$, so

$$|\varphi(v) - \varphi(w)| = ((v - w) \cdot (v - w))^{1/2} = |v - w| \quad (2.5)$$

Hence, φ is an isometry. Furthermore, since φ is a linear map, $\varphi(0) = 0$.

(ii) $\implies$ (iii): Suppose φ preserves the dot product. First, we show that φ is a linear operator on $\mathbb{R}^3$. We show that $\varphi(v + w) = \varphi(v) + \varphi(w)$ first. Consider that

$$(v + w) \cdot (v + w) = (v + w) \cdot v + (v + w) \cdot w = v \cdot v + 2(v \cdot w) + w \cdot w \quad (2.6)$$

φ is given to preserve the dot product, so

$$\begin{aligned} \varphi(v + w) \cdot \varphi(v + w) &= \varphi(v + w) \cdot \varphi(v) + \varphi(v + w) \cdot \varphi(w) \\ &= \varphi(v) \cdot \varphi(v) + 2(\varphi(v) \cdot \varphi(w)) + \varphi(w) \cdot \varphi(w) \end{aligned} \quad (2.7)$$

Therefore, it must follow that the dot products are equal:

$$\varphi(v + w) \cdot \varphi(v + w) = \varphi(v + w) \cdot (\varphi(v) + \varphi(w)) = (\varphi(v) + \varphi(w)) \cdot (\varphi(v) + \varphi(w)) \quad (2.8)$$

Let $x = \varphi(v + w)$ and $y = \varphi(v) + \varphi(w)$. Then

$$x \cdot x = x \cdot y = y \cdot y \quad (2.9)$$

This implies that

$$(x - y) \cdot (x - y) = x \cdot x - 2(x \cdot y) + y \cdot y = x \cdot x - 2(x \cdot x) + x \cdot x = 0 \quad (2.10)$$

Therefore $x - y = 0$, so $x = y$. That means that $\varphi(v + w) = \varphi(v) + \varphi(w)$. Now, we must show that $\varphi(cv) = c\varphi(v)$. Consider that

$$(cv) \cdot (cv) = c(v \cdot cv) = c^2(v \cdot v) \quad (2.11)$$

Remember, φ preserves the dot product, so

$$\varphi(cv) \cdot \varphi(cv) = c\varphi(v) \cdot \varphi(cv) = c\varphi(v) \cdot c\varphi(v) \quad (2.12)$$

It necessarily follows that $\varphi(cv) = c\varphi(v)$ by the same dot product argument given above. Therefore, φ satisfies both conditions to be a linear operator, as it is a mapping from $\mathbb{R}^3$ to $\mathbb{R}^3$. Since it also preserves the dot product, φ is orthogonal as well.

(i) $\implies$ (ii): Suppose that φ is an isometry, and $\varphi(0) = 0$. We know that

$$(\varphi(v) - \varphi(w)) \cdot (\varphi(v) - \varphi(w)) = (v - w) \cdot (v - w) \quad (2.13)$$

by the definition of an isometry and the relationship between the dot product and the norm. Let $w = 0$. Then

$$(\varphi(v) - \varphi(0)) \cdot (\varphi(v) - \varphi(0)) = (v - 0) \cdot (v - 0) \implies \varphi(v) \cdot \varphi(v) = v \cdot v \quad (2.14)$$

If $v = 0$ instead, then we find that $\varphi(w) \cdot \varphi(w) = w \cdot w$. If neither are zero, then we have

$$\varphi(v) \cdot \varphi(v) - 2(\varphi(v) \cdot \varphi(w)) + \varphi(w) \cdot \varphi(w) = v \cdot v - 2(v \cdot w) + w \cdot w \quad (2.15)$$

$$\implies v \cdot v - 2(\varphi(v) \cdot \varphi(w)) + w \cdot w = v \cdot v - 2(v \cdot w) + w \cdot w \quad (2.16)$$

$$\implies -2(\varphi(v) \cdot \varphi(w)) = -2(v \cdot w) \quad (2.17)$$

$$\implies \varphi(v) \cdot \varphi(w) = v \cdot w \quad (2.18)$$

Therefore, φ preserves dot products.

By showing that (iii) $\implies$ (i) $\implies$ (ii) $\implies$ (iii), it follows that these three conditions are equivalent. $\square$

Problem 2.2

Show that if we have an isometry f such that $f(0) = v$, then $f = t_v\varphi$, where φ is an orthogonal operator and t_v is a translation as defined before. *Hint:* Let $\varphi = t_{-v}f$.

As it turns out, the isometries form a group, as they are closed under composition, associative (mappings are associative), there is an identity, and every isometry has an inverse isometry. We also now define the group of orthogonal linear operators on $\mathbb{R}^n$, which is a subgroup of the group of isometries.

Definition 2.2 — Group of Isometries

We denote the group of isometries of $\mathbb{R}^n$ by M_n . The group of orthogonal linear operators of $\mathbb{R}^n$ is defined to be O_n .

There exists a homomorphism between M_n and O_n , as we can show below. Furthermore, this homomorphism is surjective. We construct a mapping π from M_n to O_n , and show that this mapping is a surjective homomorphism.

Proposition 2.2. Define the map $\pi : M_n \rightarrow O_n$, where, given $f = t_v\varphi$, we have $\pi(f) = \varphi$. π is a surjective homomorphism, whose kernel is the set $\{t_v \mid v \in \mathbb{R}^n\}$ of translations.

Proof. We have that π is surjective, as any orthogonal linear operator φ is also an isometry, so $\pi(\varphi) = \varphi$ for all $\varphi \in O_n$. Suppose $f = t_v\varphi$ and $g = t_w\psi$, where $\varphi, \psi \in O_n$. Then $\varphi t_w = t_{\varphi(w)}\varphi$, and

$$fg = t_v\varphi t_w\psi = t_{v+\varphi(w)}\varphi\psi \quad (2.19)$$

Therefore,

$$\pi(fg) = \varphi\psi = \pi(f)\pi(g) \quad (2.20)$$

So π is a homomorphism. Furthermore, we have that any translation can be expressed as $t_v = t_v I_n$, where I_n is the identity operator, which is in O_n . It follows that

$$\pi(t_v) = I_n \quad (2.21)$$

for all $t_v \in T \subset M_n$; it follows that T is the kernel of π . $\square$

The reader may be familiar with orientation-preserving and orientation-reversing orthogonal linear operators. We extend this definition to isometries like so.

Definition 2.3 — Orientation

We say that an **orientation-preserving** isometry $f = t_v\varphi$ is one such that $\det \varphi = 1$, where φ is an orthogonal operator on $\mathbb{R}^n$. Similarly, an isometry $f = t_v\varphi$ is **orientation-reversing** if $\det \varphi = -1$. Note that, if φ is an orthogonal operator, $\det \varphi = \pm 1$.

Let us consider the plane, $\mathbb{R}^2$. We will identify some key isometries, as well as their formulations in the language of matrices.

Definition 2.4 — Plane Isometries

There are three **plane isometries** of note:

1. **Translation:**

$$t_v(x) = x + v = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \quad (2.22)$$

2. **Rotation:**

$$\rho_\theta(x) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad (2.23)$$

3. **Reflection:**

$$r(x) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad (2.24)$$

We now come to a couple of lemmas that will assist us with the proof of an upcoming theorem that will allow us to categorize every single plane isometry. This first lemma shows that if we have a plane isometry, we must be able to write it in one of two ways, as below:

Lemma 2.3. Suppose f is a plane isometry. Then we must have either $f = t_v\rho_\theta$ (which is orientation-preserving) or $f = t_v\rho_\theta r$ (which is orientation-reversing), for some unique vector v and angle θ .

Proof. Problem 2.2 allows us to claim that we can write an isometry f as the composition of a translation and an orthogonal linear operator. Consider the matrix

$$\begin{pmatrix} \cos \theta & * \\ \sin \theta & * \end{pmatrix} \quad (2.25)$$

with $*$ unknown. The columns of an orthogonal operator must be orthonormal, with determinant ± 1 , so there are two such possibilities in the plane:

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \rho_\theta \quad (2.26)$$

and

$$S = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \rho_\theta r \quad (2.27)$$

Hence, we can write any plane isometry in either the form $t_v \rho_\theta$ or $t_v \rho_\theta r$. $\square$

To change coordinates (by an isometry) from (x, y) to (x', y') , we wish to have the relation that

$$f(x) = y \implies f'(x') = y' \quad (2.28)$$

Consider the isometry η where

$$\eta(x') = x \quad \eta(y') = y \quad (2.29)$$

It follows that

$$f' = \eta^{-1} f \eta \quad (2.30)$$

We can also show the following basic properties of plane isometries using matrices:

1. $\rho_\theta t_v = t_{\rho_\theta(v)} \rho_\theta$,
2. $r t_v = t_{r(v)} r$,
3. $r \rho_\theta = \rho_{-\theta} r$,
4. $t_v t_w = t_{v+w}$,
5. $\rho_\theta \rho_\eta = \rho_{\theta+\eta}$, and
6. $rr = 1$.

Problem 2.3

Verify the above properties using the properties of matrices.

Our second lemma allows us to show that if an isometry can be written in the form $t_v \rho_\theta$, with $\theta \neq 0$, then the isometry must be a rotation about some point in the plane.

Lemma 2.4. Suppose that an isometry $f = t_v \rho_\theta$, where $\theta \neq 0$. f is a rotation of the angle θ about some point in the plane.

Proof. We wish to show that there exists a point p such that if we translate the isometry by t_p , the new isometry f' simply becomes a rotation. We have that

$$f' = t_p^{-1} f t_p = t_p^{-1} t_v \rho_\theta t_p = t_{-p} t_v t_{\rho_\theta(p)} \rho_\theta \quad (2.31)$$

by our change of coordinate expression and by some of the basic properties of plane isometries. Therefore, this isometry amounts to a translation by the vector $w = -p + v + \rho_\theta(p) = v + \rho_\theta(p) - p$ and then a rotation by the angle θ . We want to choose such a p that $w = 0$. Consider the operator

$$I - \rho_\theta = \begin{pmatrix} 1 - \cos \theta & \sin \theta \\ -\sin \theta & 1 - \cos \theta \end{pmatrix} \quad (2.32)$$

The determinant, as one can calculate, is $2 - 2 \cos \theta$. This is only 0 when $\cos \theta = 1$, or when $\theta = 0$. We have already stated that $\theta \neq 0$, so the matrix $I - \rho_\theta$ must therefore be nonsingular, so an equation $(I - \rho_\theta)p = v$ has a solution, and so $v + \rho_\theta(p) - p = 0$ for some p . This shows that if we translate the isometry by p , the resulting isometry is simply a rotation by θ . $\square$

We come to the ultimate result of this section: a classification of all plane isometries. These isometries can take the form of a translation, a rotation, a reflection, or a glide reflection. Consider the following statement and proof below.

Theorem 2.5. *Every plane isometry must be, if orientation-preserving, a translation t_v or a rotation of an angle θ around some point, and if orientation-reversing, a reflection about some line ℓ , or a glide reflection, which is a reflection about some line ℓ followed by translation by a nonzero vector parallel to ℓ .*

Proof. We already know that an isometry can be orientation-preserving or orientation-reversing, as the determinant of any orthogonal operator is either 1 or -1 . So, suppose f is orientation-preserving, but not a translation. We need to show that f is a rotation about some point. Write $f = t_v \rho_\theta$, and $\theta \neq 0$. It follows from our previous lemma that f indeed is a rotation about some point. That settles the orientation-preserving case. Now suppose that f is orientation-reversing. We need to show that $f = t_v \rho_\theta r$ is either a glide or a reflection. Consider $\rho_\theta r$; that is a reflection about a line ℓ_0 which passes through the origin. Rotate the coordinate system so that ℓ_0 becomes the horizontal axis, and the new expression is now r . The translation t_v must still be a translation, though the coordinates of the vector may have changed, so $f' = t_v r$. Therefore,

$$f' \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = t_v \begin{pmatrix} x_1 \\ -x_2 \end{pmatrix} = \begin{pmatrix} x_1 + v_1 \\ -x_2 + v_2 \end{pmatrix} \quad (2.33)$$

This functionally acts as a reflection about the line $\ell : \{x_2 = \frac{1}{2}v_2\}$ followed by translation by the vector $\begin{pmatrix} v_1 \\ 0 \end{pmatrix}$. Furthermore, if $v_1 = 0$, then there is no translation, so f is a reflection. Therefore, if f is orientation-reversing, it may either be a reflection or a glide reflection. Hence, the theorem has been proven. $\square$

Problem 2.4

Show that, if O_2 (being the orthogonal group) is the group of isometries fixing the origin, then the group of isometries fixing a point p is the conjugate subgroup $t_p O_2 t_p^{-1}$.

2.2 Cyclic and Dihedral Groups

We are already familiar with the definition of cyclic groups. However, now, we will view them in the context of symmetries and isometries of the plane, as below.

Definition 2.5 — Cyclic Group

The **cyclic group** C_n on the plane is the group of order n generated purely by the rotation $\rho_\theta = x$, where $\theta = 2\pi/n$. In other words, C_n is generated by one element x such that $x^n = 1$.

Furthermore, we introduce a new type of group: the **dihedral group**. This group is generated by two elements, which we may call x and y , and satisfies the relations that $x^n = 1$ for some $n \in \mathbb{Z}$, $y^2 = 1$, and $yx = x^{-1}y$. Furthermore, we may see that it corresponds neatly to a group of isometries over the plane, like so:

Definition 2.6 — Dihedral Group

The **dihedral group** D_n on the plane is the group of order $2n$ generated only by two elements: the rotation $\rho_\theta = x$, where $\theta = 2\pi/n$, and the reflection $r = y$ about a line ℓ passing through the origin. In other words, the dihedral group satisfies the relations

$$x^n = 1, \quad y^2 = 1, \quad yx = x^{-1}y \quad (2.34)$$

(This third relation stems from the property of plane isometries that $r\rho_\theta = \rho_{-\theta}r$.)

The dihedral group of order $2n$ can be thought of as the set of symmetries of a regular n -sided polygon. An example with the square is detailed below.

Example: Symmetries of a Square/Regular Polygon

Consider the **square**; that is, a polygon with four sides of equal length and four congruent interior angles. One will be able to see that rotating the square by a factor of $n\pi/2$ about the center for some $n \in \{0, 1, 2, 3\}$ will cause the shape of the square to remain invariant, although the individual points may be moved to a different place. Similarly, a reflection about a line bisecting two opposite sides or about a line defined by two opposite vertices will cause the shape of the square to remain invariant. Upon further inspection, it is clear to see that the set of the possible operations that may be performed on a square such that it remains invariant, or **symmetry operations**, is indeed a group, generated by the rotation by $\pi/2$ and one of the four reflections (it could be any of them). Thus, the group of symmetries of the square is isomorphic to that of the dihedral group D_4 . Similarly, for any regular n -sided polygon, we see that the group of symmetries of such a shape is isomorphic to D_n .

Problem 2.5

For what n are D_n and S_n isomorphic? Recall that S_n is the group of permutations of $\{1, \dots, n\}$.

We now consider the idea of a **discrete subgroup** of $\mathbb{R}$. Such a subgroup can be considered discrete if every element in that group is separated from zero, the identity, by a distance of at least ϵ for some ϵ . Consequently, every element in that group must be separated from every other element by a distance of at least ϵ as well. Hence, one can adequately describe such a group as “discrete.”

Definition 2.7 — Discrete Subgroup of $\mathbb{R}$

Consider the group $\mathbb{R}$ under addition. A subgroup Γ of $\mathbb{R}$ is known to be **discrete** if there exists some $\epsilon \in \mathbb{R}$ such that for every nonzero $c \in \Gamma$, $|c| \geq \epsilon$.

Earlier on in the book, we found that every subgroup of $\mathbb{Z}$ must either be $\{0\}$ or $\mathbb{Z}a$ for some $a \in \mathbb{Z}$. We will now prove an analogous statement for discrete subgroups of $\mathbb{R}$. This will help us prove a theorem classifying subgroups of orthogonal operators on the plane, as we will see very soon.

Lemma 2.6. *All discrete subgroups Γ of $\mathbb{R}$ must either take the form of $\{0\}$ or $\mathbb{Z}a$ for some positive $a \in \mathbb{R}$.*

Proof. If $a, b \in \Gamma$, then $a - b \in \Gamma$, so $|a - b| \geq \epsilon$. Only a finite amount of elements each separated by ϵ can fit into a bounded interval, so any bounded interval must contain a finite number of elements of Γ . Note that $\{0\}$ is a discrete subgroup of $\mathbb{R}$, as it fulfills the definition vacuously. So, suppose $\Gamma \neq \{0\}$. It follows that Γ must also contain some $b \neq 0$, and by the inverse property of a group, $-b \in \Gamma$ as well. Without loss of generality, suppose $b > 0$; the group thus contains at least one positive element, and it follows that there must be some smallest positive element, since any bounded interval must contain a finite amount of elements in Γ . Call this element a . Every element za , where $z \in \mathbb{Z}$, must be in Γ , as $a + a \in \Gamma$ due to closure under addition. Therefore, $\mathbb{Z}a \subset \Gamma$. Now, suppose $b \in \Gamma$. We know $b = ra$ for some $r \in \mathbb{R}$. Suppose $r = m + r_0$, where $m \in \mathbb{Z}$ and $0 \leq r_0 < 1$. It follows that $c = b - ma$ must be in Γ by closure under addition and the inverse property. Therefore, $c = r_0a$, and $0 \leq c < a$. Recall that a is supposed to be the smallest positive element of Γ ; it follows necessarily that $c = 0 \implies r_0 = 0$, so $r \in \mathbb{Z}$. Therefore, any element $b \in \Gamma$ is also in $\mathbb{Z}a$, so $\Gamma = \mathbb{Z}a$. $\square$

Now, we come to the classification of subgroups of orthogonal operators on the plane. We shall now prove that there exist two kinds of finite subgroups of O_2 : cyclic groups and dihedral groups. We shall prove this statement below.

Theorem 2.7. *Suppose G is a finite subgroup of O_2 , the group of orthogonal operators on the plane. G must be one of the following groups for some $n \in \mathbb{Z}$:*

- i. The cyclic group C_n , or
- ii. The dihedral group D_n .

Proof. We have shown in the proof of Lemma 2.3 that any orthogonal operator on the plane can be written as either ρ_θ or $\rho_\theta r$ for some angle θ . Consider the case where all elements of G are rotations. Suppose Γ is the set of real numbers α such that $\rho_\alpha \in G$. Note that Γ is a subgroup of $\mathbb{R}$, and $2\pi \in G$. G is finite, so Γ must be discrete, and therefore $\Gamma = \mathbb{Z}\alpha$. Since $2\pi \in G$, it must be an integer multiple of α , so $\alpha = 2\pi/n$ for some $n \in \mathbb{Z}$, and therefore $G = C_n$, as one can see that G will have order n (there are n elements in Γ greater than or equal to 0 and less than 2π). Now suppose that G contains at least one reflection. Change the coordinates in such a way that $r \in G$, where r is the reflection about the horizontal axis. Suppose H is the subgroup of G consisting only of rotations in G ; it follows from the previous case that $H = C_n$. Therefore $\rho_\theta^k \in G$ and $\rho_\theta^k r \in G$ for $0 \leq k < n$. Therefore, $D_n \subset G$. Now, suppose $g \in G$. We know that g is either a rotation or a reflection. If g is a rotation, then $g \in H \subset D_n$, so g is in the dihedral group. If g is a reflection, we may be able to write it in the form $\rho_\alpha r$ for some α . Note that $r \in G$, and $g \in G$, so $gr = \rho_\alpha rr = \rho_\alpha$

must also be in G . Therefore α is some multiple of θ , where θ generates the subgroup of rotations in D_n , and therefore, $g \in D_n$. $\square$

It is now important to define an object known as the **centroid** of a set of points, which is simply taken by averaging the coordinates of all of the points, like so:

Definition 2.8 — Centroid

Suppose $S = \{s_1, \dots, s_n\}$ is a set of a finite number of points on the plane $\mathbb{R}^2$. The **centroid**, or **center of gravity**, is defined to be the point

$$p = \frac{1}{n} \sum_{i=1}^n s_i \quad (2.35)$$

with vector addition being the type of addition in this expression.

We will now show that if we have an isometry mapping a set of points to another set of points, the centroid of the first set of points would be mapped by the isometry to the centroid of the second set of points. This will be an important result in order to prove our next theorem.

Lemma 2.8. Suppose p is the centroid of $\{s_1, \dots, s_n\}$, and let f be such an isometry that $f(p) = p'$ and $f(s_i) = s'_i$. p' is the centroid of $\{s'_1, \dots, s'_n\}$.

Proof. Suppose $f = t_v$. Then $s'_i = s_i + v$ and $p' = p + v$. Thus,

$$p' = p + v = \frac{1}{n} \sum_{i=1}^n s_i + \frac{1}{n} \sum_{i=1}^n v = \frac{1}{n} \sum_{i=1}^n (s_i + v) = \frac{1}{n} \sum_{i=1}^n s'_i \quad (2.36)$$

Therefore p' is the centroid of $\{s'_1, \dots, s'_n\}$. Suppose $f = \varphi$, where φ is an orthogonal operator. By the properties of linear maps,

$$p' = \varphi(p) = \varphi \left(\frac{1}{n} \sum_{i=1}^n s_i \right) = \frac{1}{n} \sum_{i=1}^n \varphi(s_i) = \frac{1}{n} \sum_{i=1}^n s'_i \quad (2.37)$$

Therefore p' is the centroid of $\{s'_1, \dots, s'_n\}$. Otherwise, f can be written as a composition of t_v and φ , and it follows necessarily that the result p' of such a composition will also be the centroid of $\{s'_1, \dots, s'_n\}$. $\square$

We come to the final theorem of this section, the **Fixed Point Theorem**. This theorem asserts that if we have a finite group of isometries, then there must exist some point such that every isometry maps that point to itself.

Theorem 2.9 (Fixed Point Theorem). Suppose G is a finite group of plane isometries. There exists some point $p \in \mathbb{R}^2$ such that for all $g \in G$, $g(p) = p$.

Proof. Suppose $s \in \mathbb{R}^2$, and define S to be the **orbit** of s ; that is, the set of all $g(s)$ where $g \in G$. Note that $s \in S$, since $1 \in G$, and $1(s) = s$. The map $h : S \rightarrow S$, where $s \mapsto h(s)$, where $h \in G$, is a bijective map; hence, it is a permutation. Suppose $s' = g(s)$. We show that $h(s') \in S$.

$h(s') = h(g(s)) = hg(s)$, and $hg \in G$ by closure, so $hg(s) \in S$, and is therefore equal to $h(s')$. Furthermore, suppose $h(s') = h(s'')$; that is, suppose $h(g'(s)) = h(g''(s))$. Then $hg' = hg''$, which entails $g' = g''$, which implies $g'(s) = g''(s) \implies s' = s''$. Therefore, the map $h : S \rightarrow S$ is injective. Since G is finite, S is finite, so it follows that h is bijective. Therefore, any element of G will *permute* S . Consider the centroid p of S . The isometry h maps p to p' , which is the centroid of the image of $h(S)$, which is also S . Therefore $p = p'$. The centroid is therefore fixed under any element $h \in G$. Hence, there exists a point in the plane that remains fixed under every isometry $h \in G$. $\square$

Problem 2.6

Suppose G is a subgroup of M_2 , the group of plane isometries. Show that G becomes, under a coordinate change, either C_n or D_n for some $n \in \mathbb{Z}$. *Hint:* Use both theorems in this section.

2.3 Crystallographic Groups

We have discussed isometries at length, and in particular plane isometries. We will now discuss discrete groups of plane isometries, whose definition is similar to that of discrete subgroups of $\mathbb{R}$. It is required that, for a discrete group of plane isometries, all nonzero translation vectors have larger magnitude than some real number, and the absolute value of all possible rotation angles are larger than some real number as well.

Definition 2.9 — Discrete Group of Plane Isometries

We say that a group G of plane isometries is **discrete** if there exists some $\epsilon > 0$ such that if a translation by the nonzero vector v is an element of G , then $|v| \geq \epsilon$, and if a rotation by a nonzero angle θ is an element of G , then $|\theta| \geq \epsilon$.

It may be wise to distinguish between *translation vectors* and *points in the plane*, although both belong to $\mathbb{R}^2$. These objects differ in terms of their coordinate transformation properties, as one may find in the discussion below.

Remark

At this point, we should begin to think of *points* and *translation vectors* as separate entities. We say that a point $p \in \mathbb{R}^2$, whereas a translation vector $v \in V$, where V is essentially another copy of $\mathbb{R}^2$. The difference between V and $\mathbb{R}^2$ is that the origin of $\mathbb{R}^2$ may be changed, whereas the origin of V must always be the zero vector. Note that translation vectors act on points like so: $t_v(p) = p + v$. The distinction between points and translation vectors is similar to the distinction between $\mathbb{R}^2$ the *manifold* and the *tangent plane* $T_p\mathbb{R}^2 \cong \mathbb{R}^2$, but that is a story for another day, and it is not completely the same (there are copies of the tangent plane for each point, for example).

We find that for each group of plane isometries, there are two very important associated groups of note. The first we will consider is known as the **translation group**, and consists of the set of translation vectors such that the associated translation is in the group itself. The second group,

which will be discussed later, is known as the **point group**.

Definition 2.10 — Translation Group

If we have a group G of plane isometries, the **translation group** L of G is the set of vectors such that the translation by any one of those vectors is in G ; in other words,

$$L = \{v \in V \mid t_v \in G\} \quad (2.38)$$

We now consider discrete subgroups of $\mathbb{R}^n$, or perhaps the set of translation vectors V . We define such groups in the expected manner; all nonzero vectors in the group have magnitude larger than a given real number.

Definition 2.11 — Discrete Subgroup

A **discrete subgroup** is a subgroup L of either V or $\mathbb{R}^n$ such that there exists some $\epsilon > 0$ such that for all nonzero $v \in L$, $|v| \geq \epsilon$.

One particularly simple example of a discrete subgroup of $\mathbb{R}^n$ is $\mathbb{Z}^n$, the set of n -tuples of integers.

Example: $\mathbb{Z}^n$

Consider the subgroup $\mathbb{Z}^n$ of $\mathbb{R}^n$ consisting of ordered pairs of integers. Set $\epsilon = 1$. It is clear that for all $a, b \in \mathbb{Z}^n$, we have that $|a - b| \geq 1 = \epsilon$. It follows necessarily that $\mathbb{Z}^n$ is a discrete subgroup of $\mathbb{R}^n$.

Problem 2.7

Is it possible for a nontrivial translation group L of a group G of plane isometries to be finite? What about a non-discrete translation group?

Consider the following claims. If we consider a discrete subgroup of the plane, then any plane region must contain only a finite number of points in the subgroup, and if the subgroup is nontrivial, then it must contain at least one vector that is minimal by length. A proof is provided below. These results are foundational and important for future use.

Lemma 2.10. Suppose L is a discrete subgroup of V or $\mathbb{R}^2$.

- i. Any bounded plane region must contain only finitely many points in L , and
- ii. If L is not simply $\{0\}$, then it must contain some vector that is the shortest nonzero vector in L .

Proof. To begin the proof of the first claim, note that all elements of L must be separated from each other by at least a distance of ϵ . Therefore, a square with side length ϵ centered at some element of L , which may be contained within an ϵ -ball centered at that element of L , must therefore contain at most only one element in L . We say a region of the plane is **bounded** if it may be able to be contained by a ball of radius $n\epsilon$, for some $n \in \mathbb{R}$. Such a ball may be able to be sufficiently covered by a square with side length $2n\epsilon$, or, equivalently, $4n^2$ squares with side length ϵ . This implies that there are at most only $4n^2$ elements in the region, and therefore a bounded plane region must

contain only a finite number of elements of L .

To begin the proof of the second claim, suppose L is not the trivial group containing just the zero vector. There must therefore exist some nonzero $v \in L$. Consider the disk of radius $|v|$ centered at the origin; this is a bounded region of the plane. By (i), such a region will contain only finitely many nonzero elements of L . Consider the set of such elements. Since the cardinality of this set is finite, we may be able to choose a minimum (that is, a vector of minimal length) from this set. Such a vector must be minimal in length, as a smaller vector in terms of length must lie in the disk of radius $|v|$, contradicting our choice of a minimal vector. Therefore, the second claim holds true as well. $\square$

We say that $\Pi(e_1, e_2)$, where $\{e_1, e_2\}$ is a basis of $\mathbb{R}^2$, is the set of linear combinations $pe_1 + qe_2$, where $0 \leq p \leq 1$ and $0 \leq q \leq 1$. It can be thought of as the parallelogram with vertices $0, e_1, e_2, e_1 + e_2$. Let $\Pi'(e_1, e_2)$ be the set of linear combinations $pe_1 + qe_2$, where $0 \leq p < 1$ and $0 \leq q < 1$; this can be thought of as the parallelogram defined by $\Pi(e_1, e_2)$ without the edges $[e_1, e_1 + e_2]$ and $[e_2, e_1 + e_2]$. We now consider the following lemma:

Lemma 2.11. *If $\{e_1, e_2\}$ is a basis of $\mathbb{R}^2$, and L is the set of all combinations $ae_1 + be_2$, where $a, b \in \mathbb{Z}$, then every $v \in \mathbb{R}^2$ can be written uniquely as some $v = x + v_0$, where $x \in L$ and $v_0 \in \Pi'(e_1, e_2)$.*

Proof. We understand that $\{e_1, e_2\}$ is a basis of $\mathbb{R}^2$, so every vector $v \in \mathbb{R}^2$ is a linear combination of $c_1e_1 + c_2e_2$, where $c_1, c_2 \in \mathbb{R}$. Recall that every real number $r \in \mathbb{R}$ may be expressed uniquely as $r = z + r_0$, where $z \in \mathbb{Z}$ and $0 \leq r_0 < 1$. Therefore, we have $c_1 = z_1 + c_{10}$ and $c_2 = z_2 + c_{20}$, where $z_1, z_2 \in \mathbb{Z}$ and $0 \leq c_{10}, c_{20} < 1$. Write $x = z_1e_1 + z_2e_2$. Such a vector is in L . Write $v_0 = c_{10}e_1 + c_{20}e_2$. Such a vector is in $\Pi'(e_1, e_2)$. Therefore, $x + v_0$ uniquely defines v . $\square$

To facilitate further discussion, we define the **lattice**, which evokes the concept of a lattice colloquially: a repeated and “crystalline” arrangement of points. We formally define the concept below.

Definition 2.12 — Lattice

A **lattice** is a group $L = \mathbb{Z}a + \mathbb{Z}b$, where a and b are two linearly independent vectors. The set $\{a, b\}$ is called a **lattice basis** for L .

It should be noted that lattices may be higher-dimensional as well, but for the purposes of these notes, we will remain in two dimensions. We now show that any discrete subgroup of the plane must either be trivial, the set $\mathbb{Z}a$, where a is a vector in the plane, or a lattice.

Theorem 2.12. *All discrete subgroups L of V or $\mathbb{R}^2$ must be one of the following:*

- i. $L = \{0\}$,
- ii. $L = \mathbb{Z}a = \{ma \mid m \in \mathbb{Z}\}$ where a is a nonzero vector, or
- iii. $L = \mathbb{Z}a + \mathbb{Z}b = \{ma + nb \mid m, n \in \mathbb{Z}\}$, where a, b are linearly independent vectors (a **lattice**).

Proof. We know that the zero group is a discrete subgroup of V or $\mathbb{R}^2$. Suppose that $L \neq \{0\}$. Then we consider two cases:

If all elements (vectors) of L are on a line ℓ through the origin, then L is a subgroup of ℓ , which is isomorphic to $\mathbb{R}$. Since any discrete subgroup of $\mathbb{R}$ takes the form of $\mathbb{Z}a$ for some element a in the subgroup, it necessarily follows that L may be expressed as $\mathbb{Z}a$ for some element $a \in L$.

If the elements of L do not all lie on a single line through the origin, then L must contain at least two linearly independent vectors e_1 and e_2 , which therefore constitute a basis for $\mathbb{R}^2$. Consider the line ℓ spanned by e_1 . The elements of L lying on ℓ form a discrete subgroup that is not zero, so that subgroup takes the form $\mathbb{Z}a$ for some element a . If we adjust the coordinates in such a way that $a = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ in our new coordinate system, and replace $e_2 = \begin{pmatrix} e_{2,1} \\ e_{2,2} \end{pmatrix}$ with $-e_2$ only if necessary, so that $e_{2,2}$ may be positive, we aim to find a vector $b = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ such that b_2 is positive and the vector is as small as possible. We only need to consider the elements such that $0 < b_2 \leq e_{2,2}$, first of all. Additionally, since $na + b$ must be in L by closure for all $n \in \mathbb{Z}$, we can assume that $0 \leq b_1 < 1$, since otherwise, we may be able to add or subtract a multiple of a to the vector in order to make the result as small as possible. b is in a bounded region of the plane, and since L is discrete, there must be only finitely many elements in said region, so we may be able to find a minimal element b in the region. Now, we must show that $\{a, b\}$ is a lattice basis for L . Suppose we have $\tilde{L} = \mathbb{Z}a + \mathbb{Z}b$. It is clear by closure that $\tilde{L} \subset L$. Consider Lemma 2.11. As a consequence, we only need to consider vectors in $\Pi'(a, b)$ (why is that?). Suppose $c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \in L$ and $0 \leq c_1 < 1$ and $0 \leq c_2 < b_2$; in other words, c is not in $\mathbb{Z}a + \mathbb{Z}b$, unless c is the zero vector. b_2 was chosen to be the minimal positive value for the second component of any vector in L , so $c_2 = 0$. Furthermore, it follows that c must be on the line ℓ as defined earlier, which means c is an integer multiple of a , so $c = 0$. Therefore, c must be the zero vector. We can conclude by Lemma 2.11 that a vector in the group L must be in $\mathbb{Z}a + \mathbb{Z}b$, and so in this case, $L = \mathbb{Z}a + \mathbb{Z}b$. $\square$

Problem 2.8

Explain the uses of Lemma 2.11 in this proof.

We now come to the **point group**; we recall the homomorphism which assigns each element $g = t_v\varphi$ in a group of isometries its associated orthogonal linear operator φ . The point group is simply the set of all φ for the group of isometries.

Definition 2.13 — Point Group

Recall the homomorphism $\pi : M \rightarrow O_2$ such that $f \mapsto \varphi$, where $f = t_v\varphi$. Consider the restriction of this homomorphism to a discrete subgroup G :

$$\pi|_G : G \rightarrow O_2 \tag{2.39}$$

We state that the **point group** $\overline{G}$ is the image of G .

We now come to an important result: a discrete subgroup of the set of plane orthogonal operators must be finite, and furthermore, it must be either cyclic or dihedral. We mainly show that such a discrete subgroup must be finite; the second claim follows swiftly from previous results in these notes.

Proposition 2.13. *A discrete subgroup $\overline{G}$ of O_2 is finite, and thus must be either cyclic or dihedral.*

Proof. Note that every rotation in $\overline{G}$ must have a rotation angle greater than some $\epsilon > 0$, as $\overline{G}$ is discrete, so the set of rotation angles Γ must be a discrete subgroup of $\mathbb{R}$, such that $2\pi \in \Gamma$. Such a discrete subgroup can be written as $\mathbb{Z}\theta$, where $\theta = 2\pi/n$ for some $n \in \mathbb{Z}$. Therefore, since O_2 consists of rotations and reflections, there are a finite number of rotations and reflections in $\overline{G}$, so $\overline{G}$ is finite. Using Theorem 2.7, it is clear that $\overline{G}$ must either be cyclic or dihedral. $\square$

Problem 2.9

Suppose G is a discrete subgroup of the isometry group M . Suppose $a \in L$, where L is the translation group, and $\bar{g} \in \overline{G}$. Show that $\bar{g}(a) \in L$ as well. Do so by showing that $t_{\bar{g}(a)} = gt_ag^{-1}$. *Hint:* Use the property that $\phi t_v = t_{\phi(v)}\phi$ by the first and second basic properties of plane isometries, where ϕ is an orthogonal operator.

The most important theorem of this section is known as the crystallographic restriction, and serves to help classify the different discrete groups of plane isometries. We consider a subgroup of the group of symmetries of a subgroup of the plane.

Theorem 2.14 (Crystallographic Restriction). *Suppose L is a discrete subgroup of V or $\mathbb{R}^2$, and $H \subset O_2$ is a subgroup of the group of symmetries of L . If L is not the trivial group, then*

- i. *Every rotation in H must have order 1, 2, 3, 4, or 6, and*
- ii. *H is either C_n or D_n , where $n = 1, 2, 3, 4, 6$.*

Proof. Suppose $\rho_\theta \in H$ is a rotation, and $a \in L$ is nonzero, with minimal length (by Lemma 2.10). We must have that $\rho_\theta(a) \in L$ as well, and therefore $b = \rho_\theta(a) - a \in L$. Recall that a is defined to have minimal length, so $|b| \geq |a|$. Note that the vectors $a, b, \rho_\theta(a)$ form an equilateral triangle when $\theta = \pi/3$, and if $\theta < \pi/3$, then $|b| < |a|$, as θ is smaller than the other two angles in the triangle, which are of equal length as $|\rho_\theta(a)| = |a|$. Therefore, $\theta \geq \pi/3$, so H is discrete (and finite) and ρ must have order less than or equal to 6. If $\theta = 2\pi/5$, then $b' = \rho_\theta^2(a) + a$ is shorter than a , which is not possible given our choice of a as a vector of minimal length. In the coordinate system where $a = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, we have that $\rho_\theta^2(a) = \begin{pmatrix} \cos(4\pi/5) \\ \sin(4\pi/5) \end{pmatrix}$, hence $b' = \begin{pmatrix} 1 + \cos(4\pi/5) \\ \sin(4\pi/5) \end{pmatrix}$. It follows that $|b'| = \sqrt{(1 + \cos(4\pi/5))^2 + (\sin^2(4\pi/5))} \approx 0.618 < |a| = 1$. Hence, we have that $\theta \neq 2\pi/5$. Therefore, every rotation in H must have order 1, 2, 3, 4, or 6, and by Theorem 2.7, H must be either C_n or D_n for $n = 1, 2, 3, 4, 6$. $\square$

It is now time to consider what are known as crystallographic groups. The two-dimensional case is also known as a **wallpaper group**, since they describe the set of symmetries of wallpaper patterns. These groups are simply discrete groups of plane isometries with the translation group being a lattice.

Definition 2.14 — Two-Dimensional Crystallographic Group

A discrete group of plane isometries $G \subset M_2$ is known as a **two-dimensional crystallographic group** if its translation group L is a lattice.

Problem 2.10

Is it possible that an element in a point group $\overline{G}$ is not in the crystallographic group G ?

One may note that if the crystallographic group has a point group isomorphic to the cyclic group of order 4, or the dihedral group of order 4, then the lattice of the group is square.

Lemma 2.15. *Suppose a crystallographic group G has point group C_4 or D_4 , and L is the lattice of G . The lattice is square.*

Proof. Since L must be discrete, choose a vector $a \in L$ with minimal length. We have that $\bar{\rho}(a) = b \in L$, where $\bar{\rho}$ is the rotation of angle $\pi/2$ in the point group $\overline{G}$, and b is orthogonal to a . Suppose for the sake of contradiction that $\{a, b\}$ is not a lattice basis for L . By Lemma 2.11, there exists some point w of L that may be expressed as $r_1a + r_2b$, with $0 \leq r_i < 1$. Note that $|w|$ is not necessarily less than $|a|$. At least one of the points $0, a, b, a + b$ will be within the ball of radius $|a|$ centered at w , for the balls of radius $|a|$ centered at $0, a, b, a + b$ cover the region defined by $r_1a + r_2b$, with $0 \leq r_i < 1$. Label that point v . It follows that $v - w \in L$ by closure, and $|v - w| < |a|$, which contradicts the fact that $|a|$ was chosen to have minimal value. Therefore, $\{a, b\}$ must be a lattice basis for L . $\square$

Let us now example crystallographic groups with point group C_4 . We are able to define the group as the set of $t_v\rho^i$, with v in the translation group and $0 \leq i < 4$. It will be left to the reader to define crystallographic groups with point group D_4 in a similar manner.

Proposition 2.16. *Suppose G is a plane crystallographic group and its point group $\overline{G} = C_4$. Choose a coordinate system such that $L = \mathbb{Z}^2$, and $\rho = \rho_{\pi/2} \in G$. We have:*

$$G = \{t_v\rho^i \mid v \in L\} \quad (2.40)$$

where $0 \leq i < 4$.

Proof. Define G' to be the set of elements $t_v\rho^i$ such that $v \in L$. We know that $t_v \in L$, so $t_v \in G$, and $\rho \in G$, as stated before. Therefore, by closure, $t_v\rho^i \in G$. It follows shortly that $G' \subset G$. Now, suppose $g \in G$. The point group is C_4 , so no reflections or glide reflections are present in G . Therefore, $g = t_u\rho_\alpha$ for some translation vector u and angle α . The image of this element in the point group is $\bar{\rho}_\alpha$, which indicates that α is a multiple of $\pi/2$, and furthermore, $\rho_\alpha = \rho^i$ for some i . Furthermore, by closure, $g\rho^{-i} = t_u \in G$, so $u \in L$. Therefore, we know that $g \in G'$, and so $G \subset G'$. We have shown that $G = G'$, and therefore,

$$G = \{t_v\rho^i \mid v \in L\} = G' \quad (2.41)$$

$\square$

Problem 2.11

Suppose G is a plane crystallographic group with point group D_4 . If coordinates are chosen such that L has integer coordinates, and $\rho_{\pi/2} \in G$, show that G can be defined either as:

$$G = \left\{ t_v \rho_{\pi/2}^i \mid v \in L \right\} \cup \left\{ t_v \rho_{\pi/2}^i r \mid v \in L \right\} \quad (2.42)$$

or

$$G = \left\{ t_v \rho_{\pi/2}^i \mid v \in L \right\} \cup \left\{ t_u \rho_{\pi/2}^i r \mid u \in \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix} + L \right\} \quad (2.43)$$

2.4 Group Actions

A particular mathematical object of interest is that of the **group action**; in some cases, elements of groups may act as sort of mappings on another set. We make this notion precise by adding a couple key properties and defining group actions like so:

Definition 2.15 — Group Action

A **group action** is a mapping $G \times S \rightarrow S$, where we denote $(g, s) \mapsto g * s$, satisfying the following properties:

- i. We have that $1 * s = s$ for all $s \in S$, and
- ii. For all $g, g' \in G$ and $s \in S$, $(gg') * s = g * (g' * s)$, which is the **associative law**.

As one may find, sets of automorphisms as well as sets of symmetries constitute group actions for their respective sets; we discuss this further in the example box below.

Example: Groups of Automorphisms

Let F denote the set of automorphisms φ from G to G , where G is a group. Suppose that we have, for some $\varphi \in F$ and $g, g' \in G$, that $\varphi(g) = g'$. We may be able to represent this as a mapping $F \times G \rightarrow G$, where $(\varphi, g) \mapsto \varphi * g = \varphi(g) = g'$. If we consider the automorphism 1 that maps each element in G to itself, then we have that $1 * g = g$ for all $g \in G$. Furthermore, if we have two automorphisms φ, φ' , it follows that $(\varphi\varphi') * g = (\varphi \circ \varphi')(g) = \varphi(\varphi'(g)) = \varphi * (\varphi' * g)$. Both properties of a group action are satisfied, so the set of automorphisms of a group is a group action on the group G .

Problem 2.12

Show that the set of symmetries of $\mathbb{R}^2$ constitute a group action on $\mathbb{R}^2$.

When we are considering the mapping from a set to the same set induced by an element of a group, we call such a mapping “left multiplication,” as we see below.

Definition 2.16 — Left Multiplication

We say that **left multiplication** by g , where $g \in G$ and G is a group, is a mapping $m_g : S \rightarrow S$ where $m_g(s) = g * s$.

Problem 2.13

Is left multiplication by g , m_g , always bijective? Is it a permutation? Explain.

The orbit is an especially useful mathematical object; the orbit of a certain element of a set can be described as the set of all images of the element when left multiplied by an element of a group, as according to a group action. This definition is made precise below.

Definition 2.17 — Orbit

Suppose that $s \in S$. A group action of G on S induces an **orbit** of s , denoted O_s , where

$$O_s = \{s' \in S \mid \exists g \in G, s' = g * s\} \quad (2.44)$$

We supply an example of an orbit in order to assist with intuition regarding this specific object. When we consider a triangle as an element of the set of triangles in the plane, which may have a group action of the group of plane isometries, any isometry takes the triangle to a congruent triangle. Hence, the orbit of a triangle is the set of all congruent triangles in the plane.

Example: Triangles

We have that the set of isometries of $\mathbb{R}^2$ allow for a group action on S , which is the set of triangles. Consider a triangle Δ . The orbit O_Δ of the triangle Δ will be the set of all triangles congruent to Δ , as geometrically, two shapes are congruent if and only if there exists an isometry mapping one to the other.

Problem 2.14

Show that the orbits O_s of a set S partition S . *Hint:* Check if the orbits O_s are equivalence classes of S .

The second mathematical object associated with the group action is that of the **stabilizer** of an element of a set, which is the set of elements of the group such that the element remains invariant under left multiplication by the element of the group. In other words,

Definition 2.18 — Stabilizer

Let $s \in S$. The **stabilizer** G_s of s is the set

$$G_s = \{g \in G \mid gs = s\} \quad (2.45)$$

Problem 2.15

What is the stabilizer of a triangle Δ ? It is *not* the set consisting of just the identity isometry.

Below are some basic properties of the stabilizer that may be useful to note. For example, the stabilizer of the group action of an element of a group on an element of a set is a conjugate subgroup of the stabilizer of the aforementioned element of the set. The proofs are fairly straightforward.

Proposition 2.17. *Suppose S is a set and there exists a group action of G on S . Let $s \in S$ and H be the stabilizer of s . Then,*

- i. *If $a, b \in G$, then $as = bs$ if and only if $a^{-1}b \in H$ if and only if $b \in aH$.*
- ii. *If $as = s'$, then the stabilizer H' of s' is a conjugate subgroup; that is,*

$$H' = aHa^{-1} = \{g \in G \mid \exists h \in H, g = aha^{-1}\} \quad (2.46)$$

Proof. Suppose $as = bs$. This is true if and only if $a^{-1}as = a^{-1}bs$. But this is equivalent to the statement $s = a^{-1}bs$. Therefore, $a^{-1}b \in H$, as s remains invariant under the left multiplication of $a^{-1}b$. It follows that $a(a^{-1}b) \in aH \implies b \in aH$.

Regarding the second part of the proposition, suppose $g = aha^{-1}$ for some $h \in H$; then $g \in H'$. Let $as = s'$ as stated before. We have that $gs' = (aha^{-1})(as) = ah(a^{-1}a)s = ahs = as = s'$. Therefore, $aHa^{-1} \subset H'$. Since $s = a^{-1}s$, we may employ a similar argument to show that $a^{-1}H'a \subset H$, which implies $aa^{-1}H'aa^{-1} \subset aHa^{-1} \implies H' \subset aHa^{-1}$, so $H' = aHa^{-1}$. $\square$

Now we consider the relationship between the orbits and stabilizers of a group action on an element and cosets. To do so, we first define the notion of **transitivity** in the sense of a group action; the reader should have shown that the orbits of a set partition the set already. We say that a group action is transitive if the set may be partitioned into just one orbit, as seen below.

Definition 2.19 — Transitive

We say that a group action of G is **transitive** if the set S which G acts on is partitioned into just one orbit.

We now consider basic properties of G/H , the set of cosets of the subgroup H , regarding group actions and stabilizers. We observe the statements below.

Proposition 2.18. *Suppose H is a subgroup of G . Then:*

- i. *The group action of G on G/H (the set of cosets) is transitive, and*
- ii. *The stabilizer of $[H] \in G/H$ is H .*

Proof. For the first part, note that all cosets in G/H can be written as aH for some $a \in G$. We have that

$$[aH] = \{ah \mid h \in H\} \implies [aH] = a[H] \quad (2.47)$$

for some $a \in G$. Therefore, since aH is in the orbit of H , as some $a \in G$ may act on H and map it to aH , we have that the operation of G on G/H is transitive.

For the second part, we use the fact that if we consider the subgroup H , where $h \in H$, then the coset hH is equal to H . Furthermore, if $a \notin H$, then the coset aH is not equal to H , as $a \in aH$ but $a \notin H$. Therefore, the stabilizer of the coset $[H]$ is the subgroup H . $\square$

We now come to an important result. We have that there exists a bijective mapping between G/H and O_s , if H is the stabilizer of s . We further note that this bijective mapping is compatible with the group action, as the equation in the proposition indicates.

Proposition 2.19. *Suppose there exists a group action of G on S , and $s \in S$. Suppose H is the stabilizer and O_s the orbit of s . There exists a bijective mapping $\epsilon : G/H \rightarrow O_s$ where $[aH] \mapsto as$. Furthermore,*

$$\epsilon(g[C]) = g\epsilon([C]) \quad (2.48)$$

where $C \in G/H$ and $g \in G$.

Proof. We need to show that ϵ defines a legitimate mapping, so we show that if $aH = bH$, then $\epsilon(aH) = \epsilon(bH)$, or $as = bs$. Suppose $aH = bH$; then we know that $a^{-1}b \in H$, by Proposition 2.17. Furthermore, since H is the stabilizer of s as defined before, any element in H that acts on s gives s back; that is, $a^{-1}bs = s$. It quickly follows that $bs = as$. Since we may be able to follow the chain of logic backwards, as these are equivalent statements, we have that $as = bs$ implies $aH = bH$, and so ϵ is injective. Furthermore, we know that an arbitrary element of the orbit, gs , may be the image of the coset gH , so ϵ is surjective as well. Now, we show that the last relation holds. Suppose $C = aH$ for some $a \in G$. Then:

$$\epsilon(g[C]) = \epsilon(g[aH]) = \epsilon([gaH]) = gas \quad (2.49)$$

$$g\epsilon([C]) = g\epsilon([aH]) = gas \quad (2.50)$$

It follows swiftly that

$$\epsilon(g[C]) = g\epsilon([C]) \quad (2.51)$$

as desired. $\square$

2.5 The Orbit-Stabilizer Theorem

We now come to one of the most crucial results in our study of group theory, the **Orbit-Stabilizer Theorem**. This theorem takes the counting formula which we are familiar with from before and applies it to the study of group actions to determine that the order of the group acting on a set is equal to the order of the orbit of an element times the order of the stabilizer of the same element. We see the theorem statement and proof in full below:

Theorem 2.20 (Orbit-Stabilizer Theorem). *If S is a finite set on which there exists a group action of G , and G_s and O_s is the stabilizer and orbit of $s \in S$, respectively, then*

$$|G| = |G_s||O_s| \quad (2.52)$$

Proof. We know from the counting formula that if H is a subgroup of a finite group G , then

$$|G| = |H||G/H| \quad (2.53)$$

Let $H = G_s$. Then

$$|G| = |G_s||G/G_s| \quad (2.54)$$

However, we have shown in Proposition 2.19 that there exists a bijection between G/G_s and O_s , so $|O_s| = |G/G_s|$. Therefore,

$$|G| = |G_s||O_s| \quad (2.55)$$

which is the desired result. $\square$

We may apply the Orbit-Stabilizer Theorem in many unexpected ways. For example, we will be able to use the number of faces of a polyhedron to calculate the number of rotational symmetries of a polyhedron. We may also use such information to determine the number of edges and vertices. Observe the following example below, which considers the **dodecahedron**.

Example: The Dodecahedron

The dodecahedron is an interesting polyhedron with twelve faces, each of them regular pentagons. We will now use the Orbit-Stabilizer Theorem to calculate the number of rotational symmetries of the dodecahedron. Note that, since one may rotate a pentagon by multiples of $2\pi/5$ and it will remain invariant, we have that the order of the stabilizer, which is the number of rotations that leave the pentagon in the same place, so to speak, is 5. Furthermore, rotations may take the dodecahedron to any of the 12 faces, so the order of the orbit, which is the number of pentagons that a rotation may take our original pentagon to, is 12. Therefore, by the Orbit-Stabilizer Theorem, we have that the order of the group of rotational symmetries of the dodecahedron is $5 \cdot 12 = 60$.

Problem 2.16

Use the fact that the order of the group of rotational symmetries of the dodecahedron is 60 to determine the number of vertices and edges of the dodecahedron. *Hint:* What are the orders of the stabilizers and orbits for each vertex and edge? You may need to draw a diagram, or look up the shape yourself.

Problem 2.17

The **icosahedron** is a polyhedron that has 20 faces, each of them triangles. Calculate the number of rotational symmetries of the icosahedron.

2.6 Exercises

Exercises

1. Let G be the group of linear transformations of $\mathbb{R}^2$ defined by

$$r(x, y) = (-y, x), \quad s(x, y) = (x, -y) \quad (2.56)$$

G is isomorphic to a familiar group. What group am I referring to?

2. Recall the group M_2 of plane isometries, which act on the set of lines in a plane. What is the stabilizer of any given line; that is, what is the set of isometries that allows the line to remain invariant?
3. We define a **permutation representation** of a group G to be a homomorphism $\varphi : G \rightarrow S_n$. Show that there is a bijection between group actions of G on $\{1, \dots, n\}$ and permutation representations.
4. Suppose there is a group action of G on a finite S . Suppose S^g is the set of all $s \in S$ such that $gs = s$, and let G_s be the stabilizer of s .
 - (a) Show that

$$\sum_{s \in S} |G_s| = \sum_{g \in G} |S^g| \quad (2.57)$$

- (b) **Burnside's Theorem** is stated as follows:

$$|G| \cdot (\text{number of orbits}) = \sum_{g \in G} |S^g| \quad (2.58)$$

Prove this theorem.

- (c) I am making a very large friendship bracelet alone, because I have no friends. I have 100 red beads and 100 blue beads. Suppose two bracelets are "equivalent" if one can be rotated or reflected to match the other. How many different equivalence classes of bracelets are there?
5. Consider the group of rotations of $\mathbb{R}^3$, SO_3 . Show that a finite subgroup of SO_3 must be one of the following:
 - (a) The **cyclic group** C_k of rotations about a line,
 - (b) The **dihedral group** D_k of symmetries of a regular k -gon,
 - (c) The **tetrahedral group** T of rotational symmetries of a tetrahedron,
 - (d) The **octahedral group** O of rotational symmetries of a cube or octahedron, or
 - (e) The **icosahedral group** I of rotational symmetries of a dodecahedron or icosahedron.
6. Prove that any rotational symmetry of a *three-dimensional* crystal must have order 2, 3, 4, or 6.
7. How many plane crystallographic groups are there? Prove that your answer is correct. Can you classify three-dimensional crystallographic groups?

Introduction to Rings



3.1 Rings

We now shift our attention to the study of **rings**. The group is a rather simple mathematical object, and the idea of a ring is quite related to the notion of a group and furthermore gives it more complexity. We will see how this is so now.

Definition 3.1 — Ring

We say that a ring is a nonempty set R equipped with two operations $+$ and $\cdot$ in R , as well as two particular elements $0, 1 \in R$ such that the following properties hold:

- i. R with the operation $+$ and identity 0 is an abelian group,
- ii. R with the operation $\cdot$ and identity 1 is a **monoid** (that is, the operation is closed under R and associative, and there exists an identity element), and
- iii. The following laws hold for all $a, b, c \in R$:

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad (3.1)$$

$$(b + c) \cdot a = b \cdot a + c \cdot a \quad (3.2)$$

We say that R with the operation $+$ and identity 0 is called the **additive group** of the ring R , and likewise, R with the operation $\cdot$ and identity 1 is called the **multiplicative monoid** of the ring.

Further Reading

- ◇ Artin, *Algebra*, 2nd ed. An excellent algebra text used at MIT that covers very similar content as this book.
- ◇ Dummit and Foote, *Abstract Algebra*, 3rd ed. A rather comprehensive guide to introductory abstract algebra.
- ◇ Jacobson, *Basic Algebra I*, 2nd ed. Another book that these notes follow a very similar form to. Recommended by one of the mathematics professors at my university.
- ◇ Lang, *Algebra*, 3rd ed. A very similar book to Dummit and Foote in terms of comprehensiveness, yet aimed more toward the graduate level than the undergraduate level.